

# Prometheus & Grafana

## Monitoring tools:

Prometheus and Grafana

## Prometheus:

=====

Prometheus is an open source monitoring tool developed by soundcloud using GO programming language..

We can monitoring our infrastructure devices using prometheus.

Infrastructure devices means it can monitor cpu,memory,file system,network bandwidth and etc.

Monitoring: Things working as expected

Observability: why things are not working

Traces: How did a request flow though the system

How to Monitor?

We need to install prometheus in one server and install the below agents on the target machines which we want to monitor.

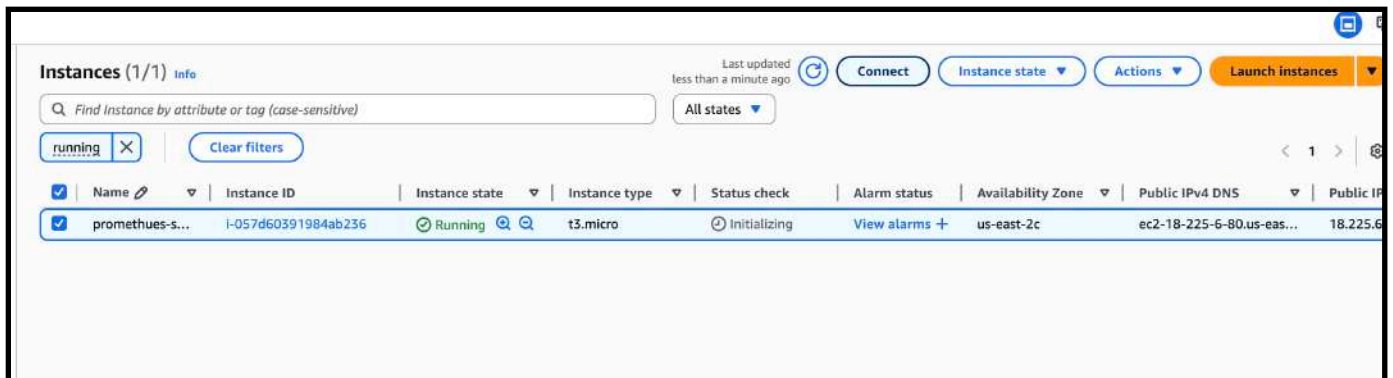
Promethues stores all the target information in prometheus.yml file.

## Tasks:

1. Install Prometheus on a server and configure prometheus.yml.
2. Install and configure Node Exporter for system monitoring.
3. Set up SNMP Exporter to monitor network devices.
4. Set up Prometheus to run as a systemd service for automatic startup.
5. Install Alertmanager and configure an email alert.
6. Install Grafana and add Prometheus as a data source.
7. Configure Prometheus for service discovery of dynamic targets.
8. Create alerting rules for monitoring disk space.
9. Create a custom Grafana dashboard using Prometheus metrics for specific insights.

## Task 1 — Install Prometheus on a server and configure prometheus.yml.

Step 1: launch one prometheus named server



Step :2 Update your server

First update packages.

```
sudo apt update / yum update
```

**Task 1 requirement:**

- ✓ Install Prometheus
- ✓ Configure `prometheus.yml`
- ✓ Verify Prometheus UI

**Step 3 — Go to /opt Directory**

```
cd /opt
```

## Download Prometheus

```
sudo wget https://github.com/prometheus/prometheus/releases/download/v3.10.0/prometheus-3.10.0.linux-amd64.tar.gz
```

```
[root@ip-172-31-46-78 opt]# sudo wget https://github.com/prometheus/prometheus/releases/download/v3.10.0/prometheus-3.10.0-linux-amd64.tar.gz
--2026-03-05 09:53:28-- https://github.com/prometheus/prometheus/releases/download/v3.10.0/prometheus-3.10.0-linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.114.3
Connecting to github.com (github.com)|140.82.114.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/191ba315-01e0-43c9-a3a3-35Z&rsct=application%2Foctet-stream&skoid=96c2deas&skt=2026-03-05T09%3A50%3A03Z&ske=2026-03-05T10%3A50%3A35Z&skb=2018-11-09&sig=AhtbGRZ%2BQ6JcdetrCLeufraNFNGaXRodWiuY29tIiwiYXVkiJoicmVsZWFZc2ZSIhc3NldHMuz210aHVidXNlcmNvbmlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjcwODAwOCwud2luZG93cy5uZUxkQ9D16xhAOH-sFUr5LzerS9R_EEyG3mz52A6dtpjR2Hfc&response-content-disposition=attachment%3B%20filename*=UTF-8''prometheus-3.10.0-linux-amd64.tar.gz following]
--2026-03-05 09:53:28-- https://release-assets.githubusercontent.com/github-production-release-asset/6838921/191ba315-01e0-43c9-a3a3-35Z&rsct=application%2Foctet-stream&skoid=96c2deas&skt=2026-03-05T09%3A50%3A03Z&ske=2026-03-05T10%3A50%3A35Z&skb=2018-11-09&sig=AhtbGRZ%2BQ6JcdetrCLeufraNFNGaXRodWiuY29tIiwiYXVkiJoicmVsZWFZc2ZSIhc3NldHMuz210aHVidXNlcmNvbmlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjcwODAwOCwud2luZG93cy5uZUxkQ9D16xhAOH-sFUr5LzerS9R_EEyG3mz52A6dtpjR2Hfc&response-content-disposition=attachment%3B%20filename*=UTF-8''prometheus-3.10.0-linux-amd64.tar.gz
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.111.133, 185.199.111.133, 185.199.111.133
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 140992264 (134M) [application/octet-stream]
Saving to: 'prometheus-3.10.0-linux-amd64.tar.gz'

prometheus-3.10.0-linux-amd64.tar.gz          100%[=====]
2026-03-05 09:53:28 (217 MB/s) - 'prometheus-3.10.0-linux-amd64.tar.gz' saved [140992264/140992264]
```

Check file :

1s

You should see:

```
prometheus-3.10.0.linux-amd64.tar.gz
```

```
[root@ip-172-31-46-78 opt]# ls
aws prometheus-3.10.0.linux-amd64.tar.gz
[root@ip-172-31-46-78 opt]# sudo tar -xvf prometheus-3.10.0.linux-amd64.tar.gz
prometheus-3.10.0.linux-amd64/
prometheus-3.10.0.linux-amd64/LICENSE
prometheus-3.10.0.linux-amd64/prometheus.yml
prometheus-3.10.0.linux-amd64/promtool
prometheus-3.10.0.linux-amd64/prometheus
prometheus-3.10.0.linux-amd64/NOTICE
[root@ip-172-31-46-78 opt]#
```

## Step 4 — Extract Prometheus

```
sudo tar -xvf prometheus-3.10.0.linux-amd64.tar.gz
```

Ls

You will see:

```
prometheus-3.10.0.linux-amd64
```

```
[root@ip-172-31-46-78 opt]# sudo tar -xvf prometheus-3.10.0.linux-amd64.tar.gz
prometheus-3.10.0.linux-amd64/
prometheus-3.10.0.linux-amd64/LICENSE
prometheus-3.10.0.linux-amd64/prometheus.yml
prometheus-3.10.0.linux-amd64/promtool
prometheus-3.10.0.linux-amd64/prometheus
prometheus-3.10.0.linux-amd64/NOTICE
[root@ip-172-31-46-78 opt]# ls
aws prometheus-3.10.0.linux-amd64 prometheus-3.10.0.linux-amd64.tar.gz
[root@ip-172-31-46-78 opt]#
```

i-057d60391984ab236 (promethues-server)

PublicIPs: 18.225.6.80 PrivateIPs: 172.31.46.78

## Step 5 — Rename Folder (Optional but Cleaner)

```
sudo mv prometheus-3.10.0.linux-amd64 prometheus
```

## Step 6 — Go Inside Prometheus Directory

```
cd prometheus
```

```
aws prometheus prometheus-3.10.0.linux-amd64.tar.gz
[root@ip-172-31-46-78 opt]# cd prometheus/
[root@ip-172-31-46-78 prometheus]# ls
LICENSE NOTICE prometheus prometheus.yml promtool
[root@ip-172-31-46-78 prometheus]#
```

## Step 7 — Check `prometheus.yml` Configuration

Open file:

```
vi prometheus.yml
```

You will see something like this:

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: "prometheus"
    static_configs:
      - targets: ["localhost:9090"]
```

Explanation:

| Setting         | Meaning                            |
|-----------------|------------------------------------|
| scrape_interval | metrics collected every 15 seconds |
| job_name        | monitoring job                     |
| targets         | which system to monitor            |

```

# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
      - targets:
        # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this config.
        labels:
          app: "prometheus"

```

Currently Prometheus is **monitoring itself**.

Save and exit:

```
:wq
```

## Step 8 — Start Prometheus

Run:

```
./prometheus --config.file=prometheus.yml
```

You will see logs like:

Server is ready to receive web requests

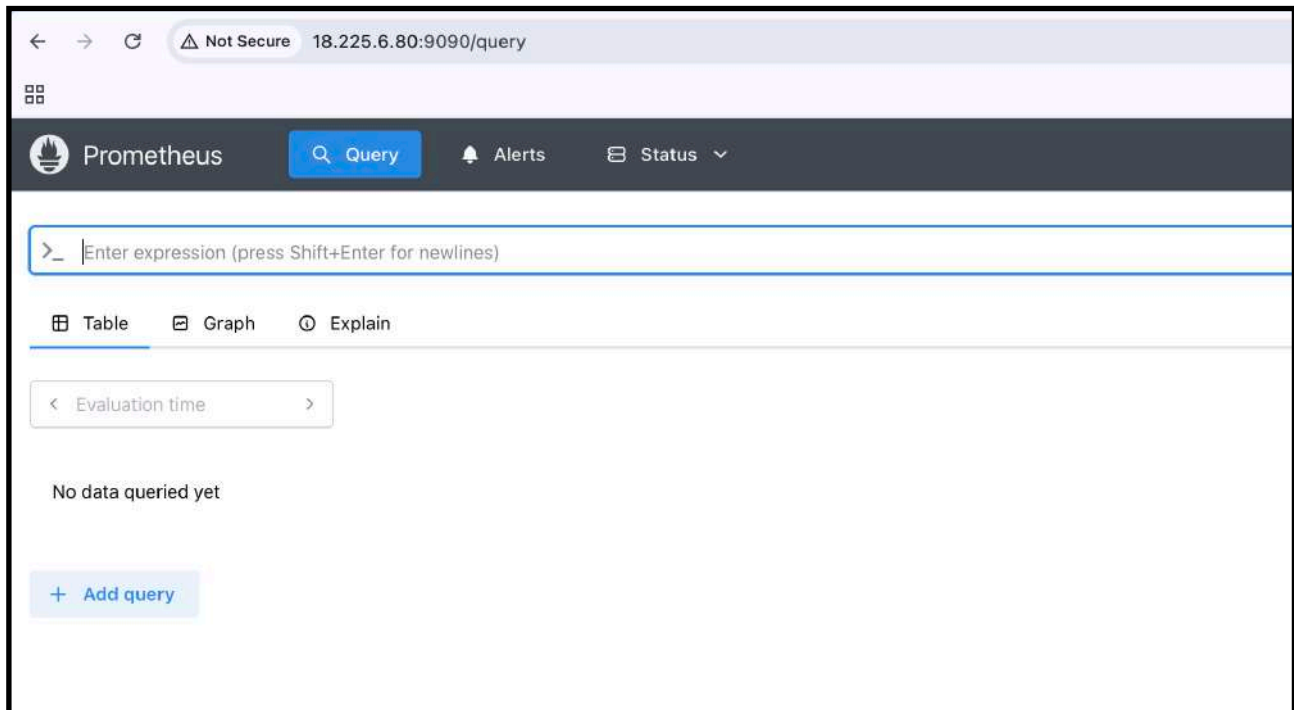
```
[root@ip-172-31-46-78 prometheus]# vi prometheus.yml
[root@ip-172-31-46-78 prometheus]# ./prometheus --config.file=prometheus.yml
time=2026-03-05T10:04:56.982Z level=INFO source=main.go:1611 msg="updated GOGC" old=100 new=75
time=2026-03-05T10:04:56.982Z level=INFO source=main.go:723 msg="Leaving GOMAXPROCS=2: CPU quota undefined" component=automaxprocs
time=2026-03-05T10:04:56.983Z level=INFO source=memlimit.go:198 msg="GOMEMLIMIT is updated" component=automemlimit package=github.com/prometheus/client_golang
44 previous=9223372036854775807
time=2026-03-05T10:04:56.983Z level=INFO source=main.go:771 msg="No time or size retention was set so using the default time retention"
time=2026-03-05T10:04:56.983Z level=INFO source=main.go:822 msg="Starting Prometheus Server" mode=server version="(version=3.1.0, branch=HEAD, buildDate=20251203-15:02:00, go.version=go1.22.0, platform=linux/amd64)"
time=2026-03-05T10:04:56.983Z level=INFO source=main.go:827 msg="operational information" build_context="(go=go1.26.0, platform=linux/amd64, tags=netgo,builtinassets)" host_details="(Linux 6.1.161-183.298.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Jan 27 05:01:22 UTC 2026 ip-172-31-46-78.ec2.internal)" fd_limits="(soft=65535, hard=65535)" vm_limits="(soft=unlimited, hard=unlimited)"
time=2026-03-05T10:04:57.154Z level=INFO source=web.go:696 msg="Start listening for connections" component=web address=0.0.0.0:9090
time=2026-03-05T10:04:57.156Z level=INFO source=main.go:1350 msg="Starting TSDB ..."
time=2026-03-05T10:04:57.158Z level=INFO source=tls_config.go:354 msg="Listening on" component=web address=[::]:9090
time=2026-03-05T10:04:57.158Z level=INFO source=tls_config.go:357 msg="TLS is disabled." component=web http2=false address=[::]:9090
time=2026-03-05T10:04:57.162Z level=INFO source=head.go:680 msg="Replaying on-disk memory mappable chunks if any" component=tsdb
time=2026-03-05T10:04:57.162Z level=INFO source=head.go:766 msg="On-disk memory mappable chunks replay completed" component=tsdb
time=2026-03-05T10:04:57.162Z level=INFO source=head.go:774 msg="Replaying WAL, this may take a while" component=tsdb
time=2026-03-05T10:04:57.163Z level=INFO source=head.go:847 msg="WAL segment loaded" component=tsdb segment=0 maxSegment=0 duration=1.476µs
time=2026-03-05T10:04:57.163Z level=INFO source=head.go:884 msg="WAL replay completed" component=tsdb checkpoint_replay_duration=897.097µs
time=2026-03-05T10:04:57.164Z level=INFO source=main.go:1371 msg="filesystem information" fs_type=XFS_SUPER_MAGIC
time=2026-03-05T10:04:57.164Z level=INFO source=main.go:1374 msg="TSDB started"
time=2026-03-05T10:04:57.164Z level=INFO source=main.go:1564 msg="Loading configuration file" filename=prometheus.yml
time=2026-03-05T10:04:57.167Z level=INFO source=main.go:1604 msg="Completed loading of configuration file" db_storage=1.421µs scrape=2.331471ms scrape_sd=32.569µs notify=120.063µs notify_sd=28.641µs rules=1.57µs tracing=4.543µs filename=prometheus.yml total_load_time=4.221µs
time=2026-03-05T10:04:57.167Z level=INFO source=main.go:1335 msg="Server is ready to receive web requests."
time=2026-03-05T10:04:57.168Z level=INFO source=manager.go:202 msg="Starting rule manager..." component="rule manager"
```

## Step 9 — Open Prometheus UI

Open browser:

`http://<EC2-PUBLIC-IP>:9090`

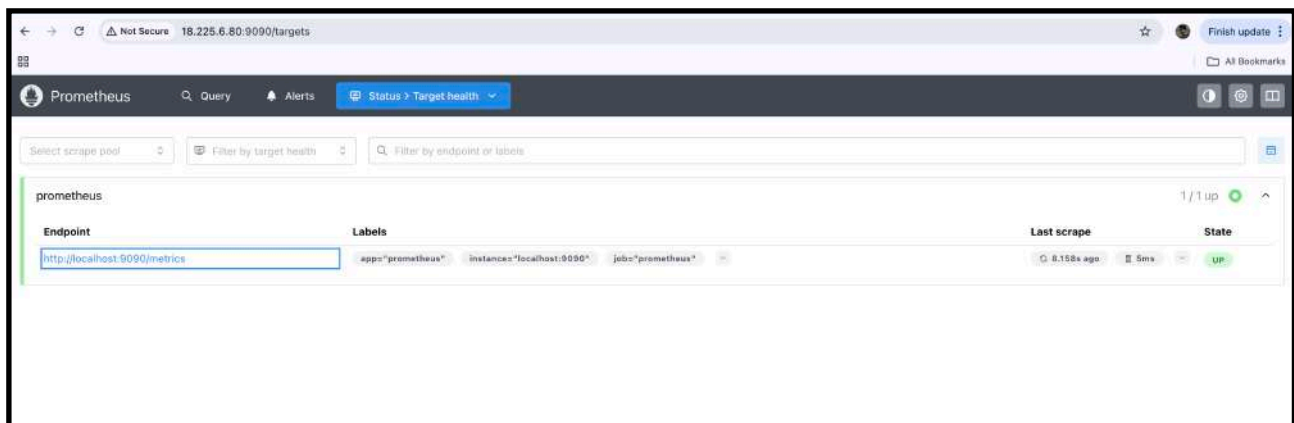




# Verify Targets

Inside Prometheus UI:

Status → Targets



This means Prometheus is successfully running.



## Task 2— Install and configure Node Exporter for system monitoring.

### Step 1 — Go to /opt

```
cd /opt
```

### Step 2 — Download Node Exporter

```
sudo wget https://github.com/prometheus/node_exporter/  
releases/download/v1.9.1/node_exporter-1.9.1.linux-  
amd64.tar.gz
```

### Extract the Package

```
sudo tar -xvf node_exporter-1.9.1.linux-amd64.tar.gz
```

Rename folder:

```
sudo mv node_exporter-1.9.1.linux-amd64 node_exporter
```

```

root@ip-172-31-46-78 opt]# sudo wget https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
--2026-03-05 10:15:52-- https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.114.3
Connecting to github.com (github.com)|140.82.114.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/dc8ec09c-2975-42a2-9591-57dd1ffff7b7?sp=r&sv=2018-1
3A17&rsct=attachment%3B+filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7
b4de&skt=2026-03-05T09%3A52%3A56Z&ske=2026-03-05T10%3A53%3A17Z&sks=b&skv=2018-11-09&sig=EzJ2M3%2F3deIyQ14krcWpMBnk%2FDwF%2BJv4MsFvv40c4TM%3D&jwt=e
9iOiJnaXRodWIuY29tIiwiaXVkiJoicmVsZWZzSlhc3NldHMuz210aHvidXNlcmNvbnRlbnQuY29tIiwia2V5MSIsImV4cCI6MTc3MjcwNzQ0NiwiImJmIjoxNzcyNzAlNjQ2LCJw
9iLmNvcuUud2luZG93cy5uZXQifQ.hws3gT2TYAmsda6DSkShJ9ZiPlAmvFLUR_9x6dWa8h8&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.9
1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-03-05 10:15:52-- https://release-assets.githubusercontent.com/github-production-release-asset/9524057/dc8ec09c-2975-42a2-9591-57dd1ffff7b7
-03-05T10%3A53%3A17Z&rsct=attachment%3B+filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-
9-b12b-9515b896b4de&skt=2026-03-05T09%3A52%3A56Z&ske=2026-03-05T10%3A53%3A17Z&sks=b&skv=2018-11-09&sig=EzJ2M3%2F3deIyQ14krcWpMBnk%2FDwF%2BJv4MsFvv
40c4TM%3D&jwt=e9iOiJnaXRodWIuY29tIiwiaXVkiJoicmVsZWZzSlhc3NldHMuz210aHvidXNlcmNvbnRlbnQuY29tIiwia2V5MSIsImV4cCI6MTc3MjcwNzQ0NiwiImJmIjoxNzcyNzAlNjQ2LCJw
9iLmNvcuUud2luZG93cy5uZXQifQ.hws3gT2TYAmsda6DSkShJ9ZiPlAmvFLUR_9x6dWa8h8&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.9
1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11582410 (11M) [application/octet-stream]
Saving to: 'node_exporter-1.9.1.linux-amd64.tar.gz'

node_exporter-1.9.1.linux-amd64.tar.gz 100%[=====]
2026-03-05 10:15:52 (162 MB/s) - 'node_exporter-1.9.1.linux-amd64.tar.gz' saved [11582410/11582410]

root@ip-172-31-46-78 opt]# ls
aws node_exporter-1.9.1.linux-amd64.tar.gz prometheus prometheus-3.10.0.linux-amd64.tar.gz
root@ip-172-31-46-78 opt]# tat xvf node_exporter-1.9.1.linux-amd64.tar.gz
-bash: tat: command not found
root@ip-172-31-46-78 opt]# tar xvf node_exporter-1.9.1.linux-amd64.tar.gz
node_exporter-1.9.1.linux-amd64/
node_exporter-1.9.1.linux-amd64/LICENSE
node_exporter-1.9.1.linux-amd64/NOTICE
node_exporter-1.9.1.linux-amd64/node_exporter
root@ip-172-31-46-78 opt]# sudo mv node_exporter-1.9.1.linux-amd64 node_exporter
root@ip-172-31-46-78 opt]# ls
aws node_exporter node_exporter-1.9.1.linux-amd64.tar.gz prometheus prometheus-3.10.0.linux-amd64.tar.gz
root@ip-172-31-46-78 opt]#

```

## Step 4 — Test Node Exporter

Go inside the folder.

```
cd node_exporter
```

Run it:

```
./node_exporter
```

You should see:

```
Listening on :9100
```



```
← → ↻ ⚠ Not Secure 18.225.6.80:9100/metrics

# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC env
/gc/gogc:percent
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise math.MaxInt64. This value is set by th
from /gc/gomemlimit:bytes
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 7
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.23.7"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated in heap and currently in use. Equals to /memory/classes/heap/objects:
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 868800
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated in heap until now, even if released already. Equals to /g
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 868800
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table. Equals to /memory/classes/prof
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.445432e+06
# HELP go_memstats_frees_total Total number of heap objects frees. Equals to /gc/heap/frees:objects + /gc/heap/tiny/allocs:obj
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 770
```

Now stop it:

CTRL + C

## Step 6 — Create Node Exporter Service

Create service file.

```
sudo vi /etc/systemd/system/node_exporter.service
```

Paste this:

```
[Unit]
Description=Node Exporter
After=network.target

[Service]
User=nobody
ExecStart=/opt/node_exporter/node_exporter
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

```
[Unit]
Description=Node Exporter
After=network.target

[Service]
User=nobody
ExecStart=/opt/node_exporter/node_exporter
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

## Step 7 — Start Service

Reload systemd:

```
sudo systemctl daemon-reload
```

Start service:

```
sudo systemctl start node_exporter
```

Enable auto start:

```
sudo systemctl enable node_exporter
```

```
C
[root@ip-172-31-46-78 node_exporter]# sudo vi /etc/systemd/system/node_exporter.service
[root@ip-172-31-46-78 node_exporter]# sudo systemctl daemon-reload
[root@ip-172-31-46-78 node_exporter]# sudo systemctl start node_exporter
[root@ip-172-31-46-78 node_exporter]# sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.s
```



## Step 8 — Verify Service

```
sudo systemctl status node_exporter
```

```
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.
[root@ip-172-31-46-78 node_exporter]# sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 10:24:07 UTC; 10s ago
     Main PID: 3222 (node_exporter)
        Tasks: 4 (limit: 1067)
       Memory: 2.1M
          CPU: 10ms
      CGroup: /system.slice/node_exporter.service
              └─3222 /opt/node_exporter/node_exporter

Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:07 ip-172-31-46-78.us-east-2.compute.internal node_exporter[3222]: time=2026-03-05T10:07:00.000Z
Mar 05 10:24:11 ip-172-31-46-78.us-east-2.compute.internal systemd[1]: [ ] /etc/systemd/system/node_exporter.service
lines 1-20/20 (END)
```

## Step 9 — Add Node Exporter to Prometheus

Edit Prometheus config.

```
sudo vi /opt/prometheus/prometheus.yml
```

Add this:

```
- job_name: 'node_exporter'
  static_configs:
    - targets: ['localhost:9100']
```

```
# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label `label_name=<label_value>` to any times
        labels:
          app: "prometheus"

- job_name: "node_exporter"
  static_configs:
    - targets: ["localhost:9100"]
```

Save exit :

## Step 10 — Restart Prometheus (Manual Run)

Since Prometheus service is **not created yet**, run it manually:

```
cd /opt/prometheus
./prometheus --config.file=prometheus.yml
```

## Step 11 — Verify Monitoring

Open browser:

<http://<EC2-PUBLIC-IP>:9090>

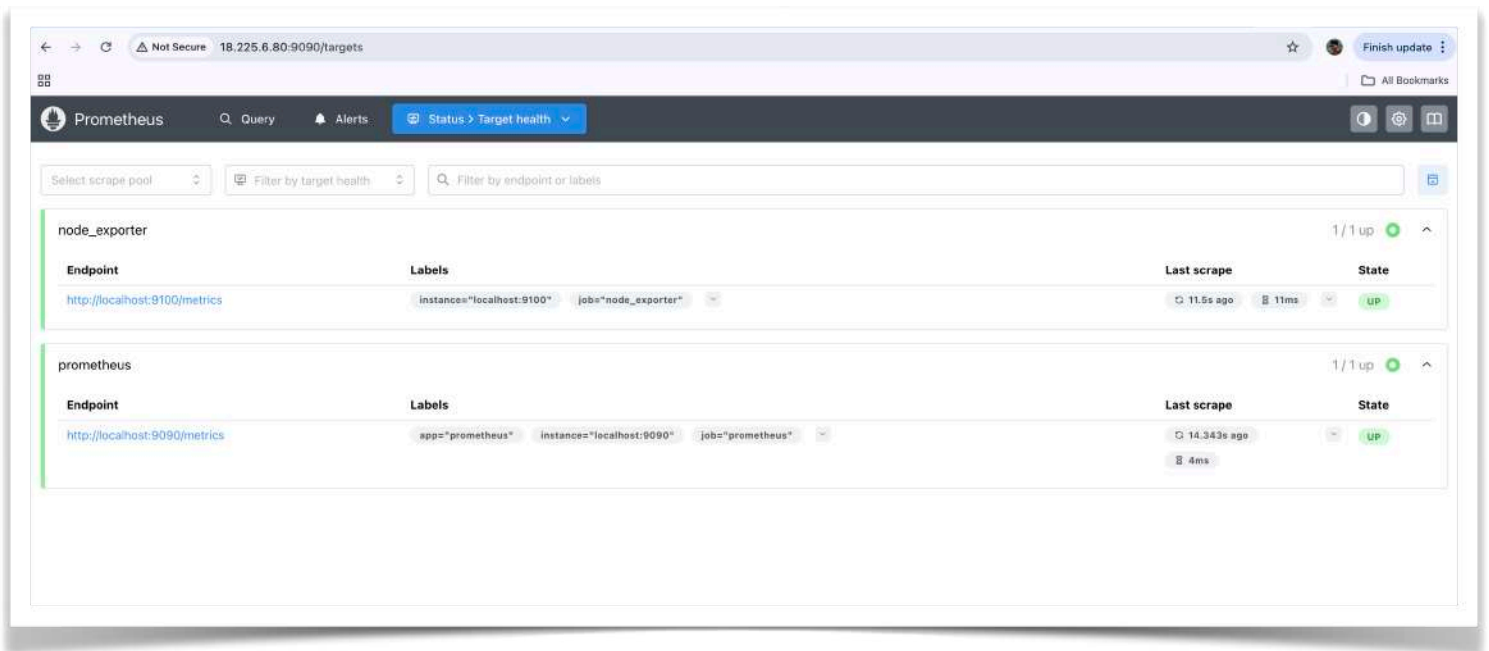
Go to:



Status → Targets

You should see:

```
localhost:9090    UP
localhost:9100    UP
```



## Task 2 Result

Prometheus is now collecting **system metrics from Node Exporter**, including:

- CPU usage
- Memory usage
- Disk usage
- Network bandwidth

## Task 3 –Set up SNMP Exporter to monitor network devices.

We will install **SNMP Exporter** on the same **Prometheus** server.

## Step 1 — Go to /opt

```
cd /opt
```

## Step 2 — Download SNMP Exporter

```
wget https://github.com/prometheus/snmp_exporter/releases/download/v0.26.0/snmp_exporter-0.26.0.linux-amd64.tar.gz
```

Check file:

1s

You should see:

```
snmp exporter-0.26.0.linux-amd64.tar.gz
```

```
[root@ip-172-31-46-78 ~]# cd /opt
[root@ip-172-31-46-78 ~]# wget https://github.com/prometheus/snmp_exporter/releases/download/v0.26.0/snmp_exporter-0.26.0.linux-amd64.tar.gz
--2026-03-05 10:46:34-- https://github.com/prometheus/snmp_exporter/releases/download/v0.26.0/snmp_exporter-0.26.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)[140.82.112.3]:443... connected.
HTTP request sent, awaiting response... 302 Found
location: https://release-assets.githubusercontent.com/github-production-release-asset/31185891/f7a2a981-2bdb-4f61-ale9-d153651ec43f?sp=r&sv=2018-11-3A41Z&rscd=attachment%3Bfilename%3Dsnmp_exporter-0.26.0.linux-amd64.tar.gz&rsc=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7a96b4de&ssk=2026-03-05T11%3A23%3A41Z&sks=2026-03-05T11%3A23%3A41Z&sks=b&skv=2018-11-09&sig=j7iVudn8lJZal6ERvwHkYVP%2BCN8lIGeGt2BRHImiqYt2F1c%3D&jwt=eysCMioIjNaXroDWiuZt9t1wiYXVKtjoicmVsZFwZS1hc3NldHMuz210aHVidNclmVnbmlRbmQyZ9t1wia2V5Ijoia2V5MSIsImV4cCtGMtc3MjcWZScWibmJmJjoxNzcyZnZAZ3NTkOLCJwJmVuc9ImlmNmVmUdd2luZG93cy5uZHQxqfQ.._tRxPsSuWLoK9MXsbdb36j5Jndj5zyUMyyfvftIROjMI&response-content-disposition=attachment%3B%20filename%3Dsnmp_exporter-0.26.0.type=application%2Foctet-stream [following]
--2026-03-05 10:46:35-- https://release-assets.githubusercontent.com/github-production-release-asset/31185891/f7a2a981-2bdb-4f61-ale9-d153651ec43f?sp=r&sv=2018-11-3A41Z&rscd=attachment%3Bfilename%3Dsnmp_exporter-0.26.0.linux-amd64.tar.gz&rsc=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7a96b4de&ssk=2026-03-05T11%3A23%3A41Z&sks=2026-03-05T11%3A23%3A41Z&sks=b&skv=2018-11-09&sig=j7iVudn8lJZal6ERvwHkYVP%2BCN8lIGeGt2BRHImiqYt2F1c%3D&jwt=eysCMioIjNaXroDWiuZt9t1wiYXVKtjoicmVsZFwZS1hc3NldHMuz210aHVidNclmVnbmlRbmQyZ9t1wia2V5Ijoia2V5MSIsImV4cCtGMtc3MjcWZScWibmJmJjoxNzcyZnZAZ3NTkOLCJwJmVuc9ImlmNmVmUdd2luZG93cy5uZHQxqfQ.._tRxPsSuWLoK9MXsbdb36j5Jndj5zyUMyyfvftIROjMI&response-content-disposition=attachment%3B%20filename%3Dsnmp_exporter-0.26.0.type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 8740509 (8.3M) [application/octet-stream]
Saving to: 'snmp_exporter-0.26.0.linux-amd64.tar.gz'

snmp_exporter-0.26.0.linux-amd64.tar.gz      100%[=====]
2026-03-05 10:46:35 (252 MB/s) - 'snmp_exporter-0.26.0.linux-amd64.tar.gz' saved [8740509/8740509]

[root@ip-172-31-46-78 ~]# s
-bash: s: command not found
[root@ip-172-31-46-78 ~]# ls
aws node exporter node exporter-1.9.1.linux-amd64.tar.gz prometheus prometheus-3.10.0.linux-amd64.tar.gz snmp exporter-0.26.0.linux-amd64.tar.gz
```

## Step 3 — Extract Package

```
tar -xvf snmp_exporter-0.26.0.linux-amd64.tar.gz
```

Rename folder:

```
mv snmp_exporter-0.26.0.linux-amd64 snmp_exporter
```

## Step 4 — Go Inside Folder

```
cd snmp_exporter
```

Check files:

```
ls
```

You should see:

```
snmp_exporter
snmp.yml
```

```
aws node_exporter node_exporter-1.9.1.linux-amd64.tar.gz prometheus prometheus-3.10.0.linux-amd64.tar.gz snmp_exporter
[root@ip-172-31-46-78 opt]# tar -xvf snmp_exporter-0.26.0.linux-amd64.tar.gz
snmp_exporter-0.26.0.linux-amd64/
snmp_exporter-0.26.0.linux-amd64/NOTICE
snmp_exporter-0.26.0.linux-amd64/snmp.yml
snmp_exporter-0.26.0.linux-amd64/LICENSE
snmp_exporter-0.26.0.linux-amd64/snmp_exporter
[root@ip-172-31-46-78 opt]# mv snmp_exporter-0.26.0.linux-amd64 snmp_exporter
[root@ip-172-31-46-78 opt]# ls
aws node_exporter node_exporter-1.9.1.linux-amd64.tar.gz prometheus prometheus-3.10.0.linux-amd64.tar.gz snmp_exporter
[root@ip-172-31-46-78 opt]# cd snmp_exporter
[root@ip-172-31-46-78 snmp_exporter]# ls
LICENSE NOTICE snmp.yml snmp_exporter
[root@ip-172-31-46-78 snmp_exporter]#
```

i-057d60391984ab236 (promethues-server)

## Step 5 — Start SNMP Exporter

```
./snmp_exporter
```

You should see something like:

```
Listening on :9116
```

```
LICENSE NOTICE snmp.yml snmp_exporter
[root@ip-172-31-46-78 snmp_exporter]# ./snmp_exporter
ts=2026-03-05T10:49:48.441Z caller=main.go:212 level=info msg="Starting snmp_exporter" version="(version=0.26.0, branch=HEAD, revision=44f8732988e726bad3f13d5779f1da7705178642)"
ts=2026-03-05T10:49:48.442Z caller=main.go:213 level=info build_context="(go=go1.22.3, platform=linux/amd64, compiler=gc, platform_toolchain=linux)"
ts=2026-03-05T10:49:48.446Z caller=tls_config.go:313 level=info msg="Listening on" address=":::9116"
ts=2026-03-05T10:49:48.546Z caller=tls_config.go:316 level=info msg="TLS is disabled." http2=false address=":::9116"
```

i-057d60391984ab236 (promethues-server)

PublicIPs: 18.225.6.80 PrivateIPs: 172.31.46.78

## Step 6 — Verify Exporter

Open browser:

```
http://<EC2-PUBLIC-IP>:9116
```

← → ↻ ⚠ Not Secure 18.225.6.80:9116



### SNMP Exporter

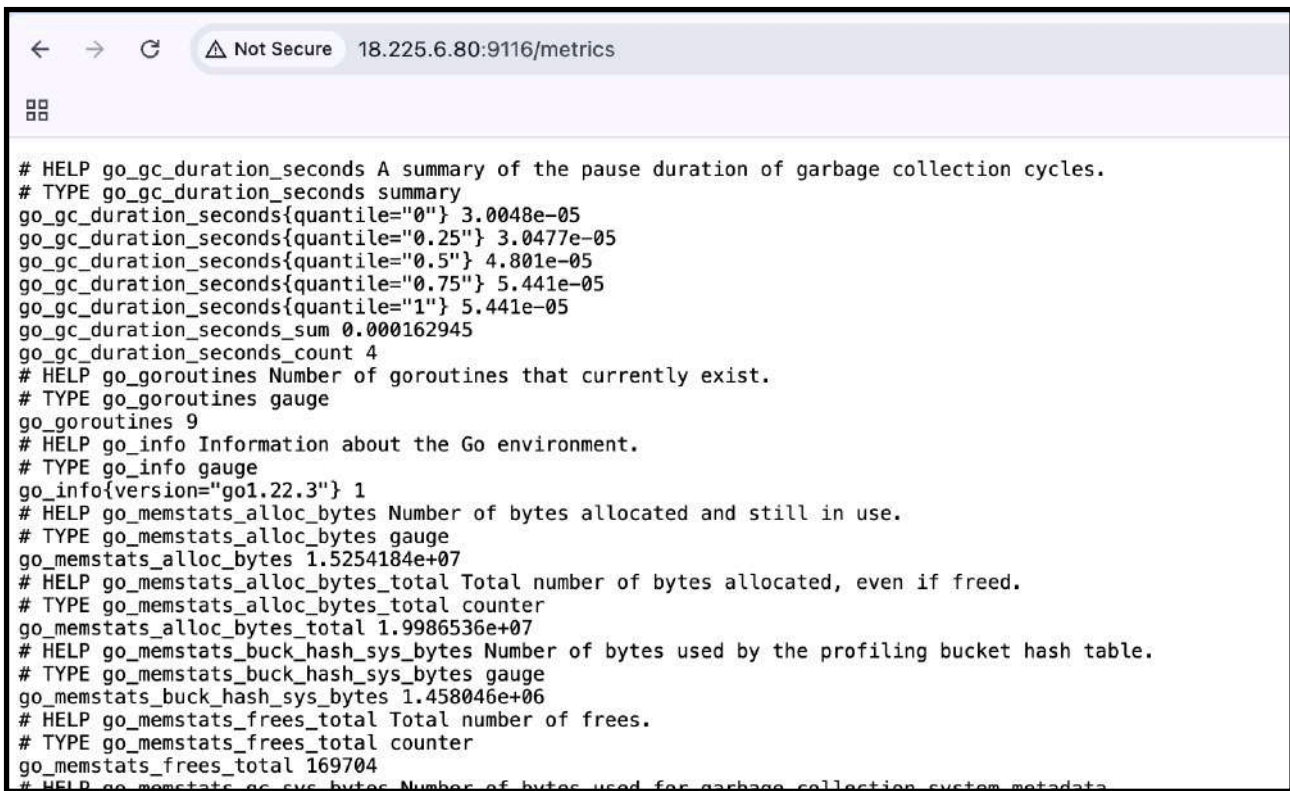
#### Prometheus Exporter for SNMP targets

Version: (version=0.26.0, branch=HEAD, revision=44f8732988e726bad3f13d5779f1da7705178642)

- [Config](#)
- [Metrics](#)

Target:   
Auth:   
Module:

Check metric



```
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 3.0048e-05
go_gc_duration_seconds{quantile="0.25"} 3.0477e-05
go_gc_duration_seconds{quantile="0.5"} 4.801e-05
go_gc_duration_seconds{quantile="0.75"} 5.441e-05
go_gc_duration_seconds{quantile="1"} 5.441e-05
go_gc_duration_seconds_sum 0.000162945
go_gc_duration_seconds_count 4
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 9
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.22.3"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated and still in use.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 1.5254184e+07
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated, even if freed.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 1.9986536e+07
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.458046e+06
# HELP go_memstats_frees_total Total number of frees.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 169704
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
```

Stop it:

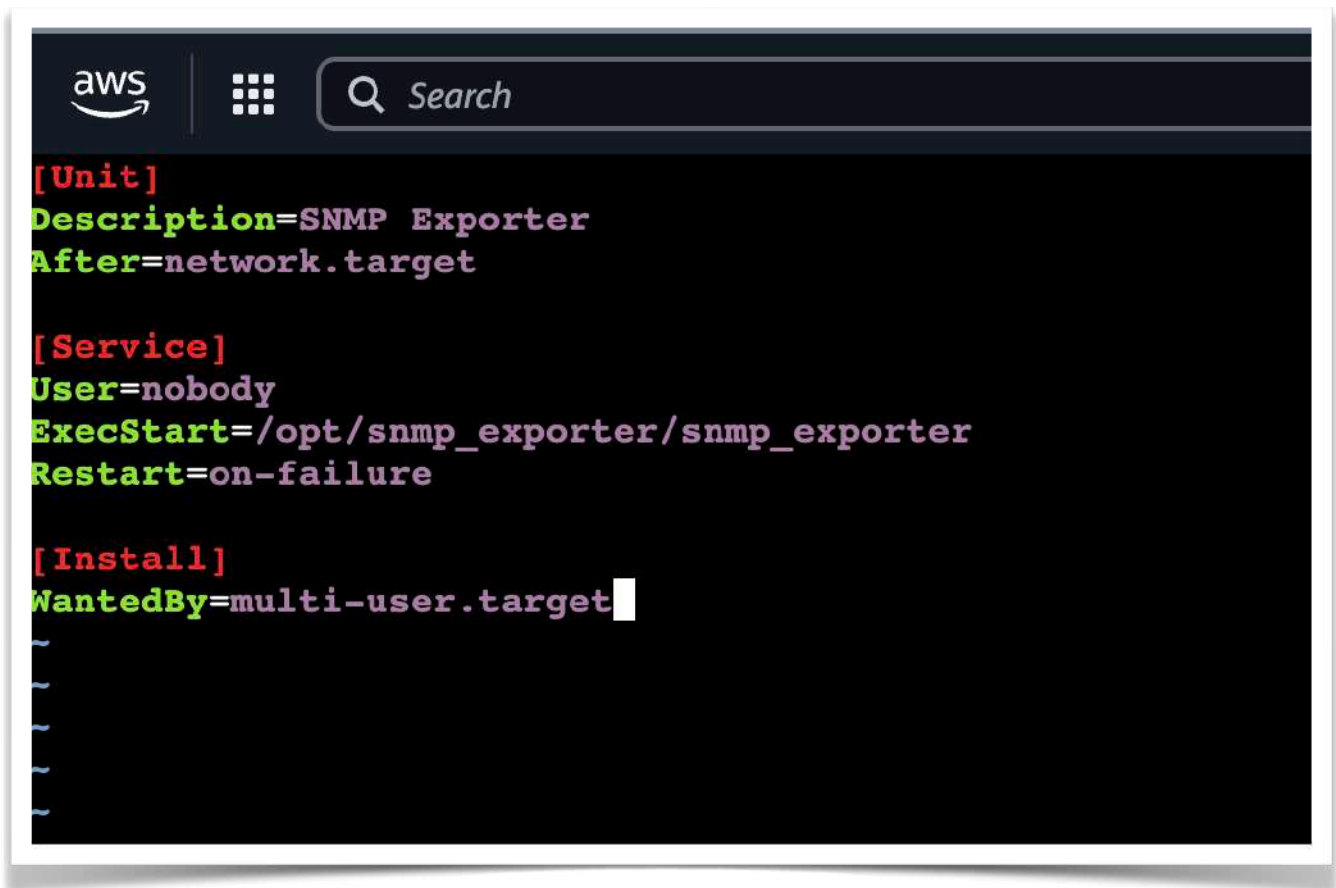
CTRL + C

## Step 7 — Create SNMP Exporter Service

Create service file.

```
vi /etc/systemd/system/snmp_exporter.service
```

Paste this:



## Step 8 — Reload systemd

```
systemctl daemon-reload
```

## Step 9 — Start SNMP Exporter

```
systemctl start snmp_exporter
```

Enable auto start:

```
systemctl enable snmp_exporter
```

## Step 10 — Verify Service

```
systemctl status snmp_exporter
```



```

root@ip-172-31-46-78 snmp_exporter]# vi /etc/systemd/system/snmp_exporter.service
root@ip-172-31-46-78 snmp_exporter]# systemctl daemon-reload
root@ip-172-31-46-78 snmp_exporter]# systemctl start snmp_exporter
root@ip-172-31-46-78 snmp_exporter]# systemctl enable snmp_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/snmp_exporter.service → /etc/systemd/system/snmp_exporter.service.
root@ip-172-31-46-78 snmp_exporter]# systemctl status snmp_exporter
● snmp_exporter.service - SNMP Exporter
   Loaded: loaded (/etc/systemd/system/snmp_exporter.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 10:53:17 UTC; 9s ago
 Main PID: 4255 (snmp_exporter)
    Tasks: 6 (limit: 1067)
   Memory: 1.6M
      CPU: 6ms
   CGroup: /system.slice/snmp_exporter.service
           └─4255 /opt/snmp_exporter/snmp_exporter

Mar 05 10:53:17 ip-172-31-46-78.us-east-2.compute.internal systemd[1]: Started snmp_exporter.service - SNMP Exporter.
Mar 05 10:53:17 ip-172-31-46-78.us-east-2.compute.internal snmp_exporter[4255]: ts=2026-03-05T10:53:17.775Z caller=main.go:110 msg="Starting snmp_exporter"
Mar 05 10:53:17 ip-172-31-46-78.us-east-2.compute.internal snmp_exporter[4255]: ts=2026-03-05T10:53:17.775Z caller=main.go:110 msg="Starting snmp_exporter"
Mar 05 10:53:17 ip-172-31-46-78.us-east-2.compute.internal snmp_exporter[4255]: ts=2026-03-05T10:53:17.776Z caller=main.go:110 msg="Starting snmp_exporter"
Mar 05 10:53:22 ip-172-31-46-78.us-east-2.compute.internal systemd[1]: /etc/systemd/system/snmp_exporter.service: line 1-16/16 (END)

```

i-057d60391984ab236 (promethues-server)

## Step 11 — Add SNMP Exporter in Prometheus

Edit config:

```
vi /opt/prometheus/prometheus.yml
```

Add this job **below** `node_exporter`:

```
- job_name: "snmp"
  static_configs:
    - targets: ["localhost:9116"]
```

```

# Here is a Prometheus job configuration
scrape_configs:
  # The job name is added as a label `job=<job_name>` to the timeseries
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label `label_name`
        labels:
          app: "prometheus"

  - job_name: "node_exporter"
    static_configs:
      - targets: ["localhost:9100"]
  - job_name: "snmp"
    static_configs:
      - targets: ["localhost:9116"]

```

# Step 12 — Restart Prometheus

Since service is not created yet, run manually:

```
cd /opt/prometheus
./prometheus --config.file=prometheus.yml
```

## Verify

Open:

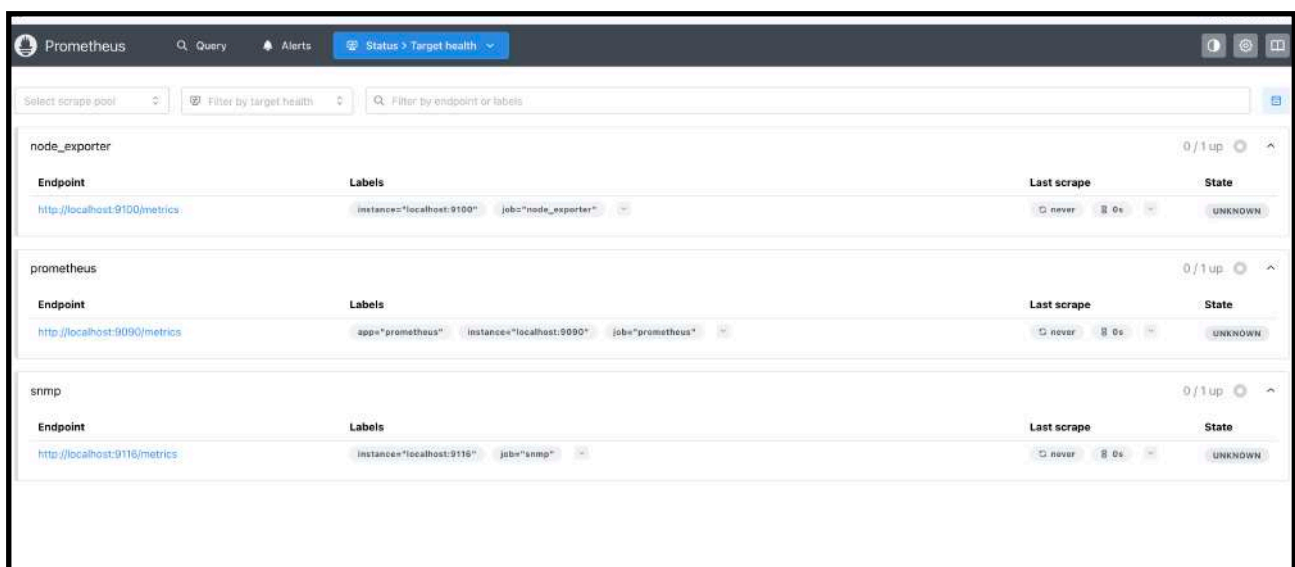
`http://<EC2-IP>:9090`

Then go to:

Status → Targets

You should see:

```
localhost:9090    UP
localhost:9100    UP
localhost:9116    UP
```



## Task 4 — Set up Prometheus to run as a systemd service for automatic startup.

### Set up Prometheus to Run as a systemd Service

## Step 1 — Create Data Directory

Prometheus stores metrics here.

```
mkdir -p /opt/prometheus/data
```

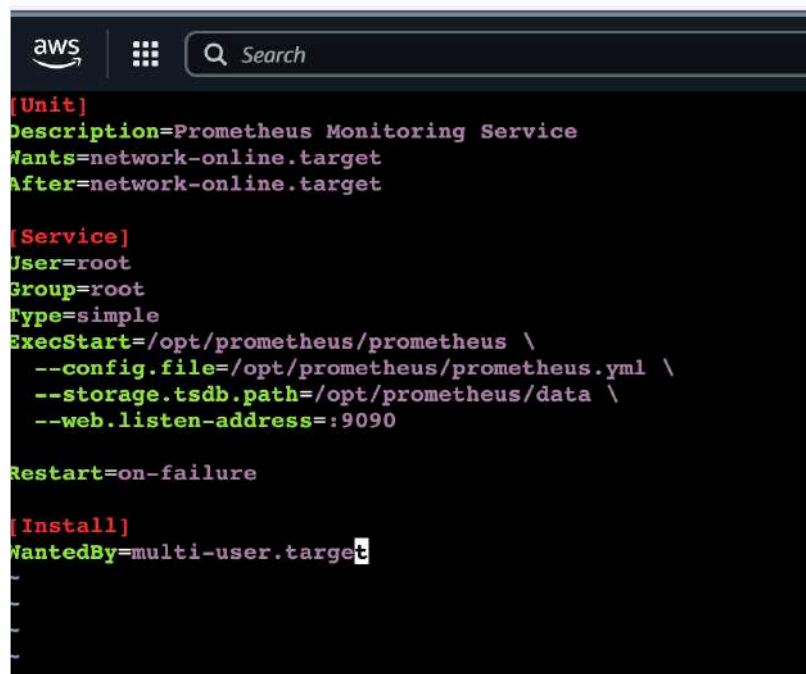
## Step 2 — Create Prometheus Service File

Create the service file.

```
vi /etc/systemd/system/prometheus.service
```

## Step 3 — Add the Service Configuration

Paste this inside the file.

A screenshot of a terminal window with a dark background. The terminal shows the content of the Prometheus service file. At the top, there is an AWS logo and a search bar. The configuration is as follows:

```
[Unit]
Description=Prometheus Monitoring Service
Wants=network-online.target
After=network-online.target

[Service]
User=root
Group=root
Type=simple
ExecStart=/opt/prometheus/prometheus \
  --config.file=/opt/prometheus/prometheus.yml \
  --storage.tsdb.path=/opt/prometheus/data \
  --web.listen-address=:9090

Restart=on-failure

[Install]
WantedBy=multi-user.target
```

This allows you to manage Prometheus using:

```
systemctl start prometheus
systemctl stop prometheus
systemctl restart prometheus
```

Now save exit

```
systemctl daemon-reload
systemctl start prometheus
```

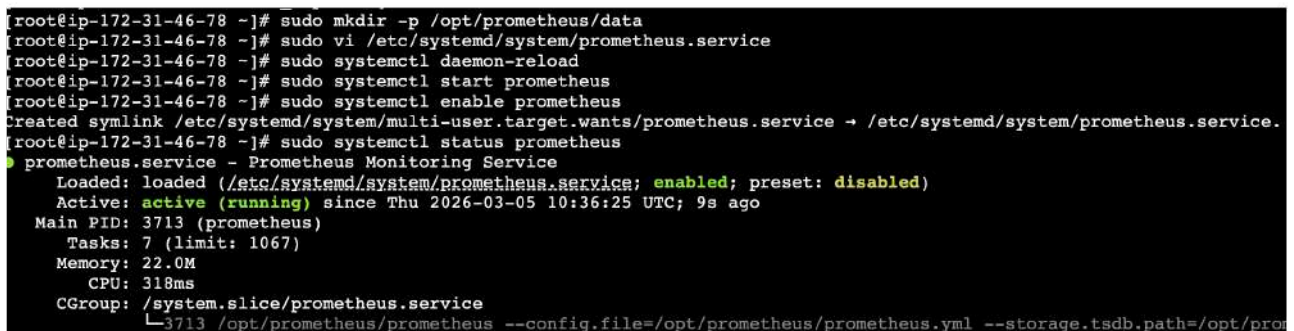
## Step 4 — Enable Service at Boot

```
systemctl enable prometheus
```

This ensures Prometheus starts **automatically after reboot**.

## Step 5 — Verify Service

```
systemctl status prometheus
```



```
root@ip-172-31-46-78 ~]# sudo mkdir -p /opt/prometheus/data
root@ip-172-31-46-78 ~]# sudo vi /etc/systemd/system/prometheus.service
root@ip-172-31-46-78 ~]# sudo systemctl daemon-reload
root@ip-172-31-46-78 ~]# sudo systemctl start prometheus
root@ip-172-31-46-78 ~]# sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service -> /etc/systemd/system/prometheus.service.
root@ip-172-31-46-78 ~]# sudo systemctl status prometheus
● prometheus.service - Prometheus Monitoring Service
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 10:36:25 UTC; 9s ago
     Main PID: 3713 (prometheus)
       Tasks: 7 (limit: 1067)
      Memory: 22.0M
         CPU: 318ms
    CGroup: /system.slice/prometheus.service
            └─3713 /opt/prometheus/prometheus --config.file=/opt/prometheus/prometheus.yml --storage.tsdb.path=/opt/prom
```

## Step 6 — Test Service Commands

Stop service:

```
systemctl stop prometheus
```

Start service:

```
systemctl start prometheus
```

Restart service:

```
systemctl restart prometheus
```

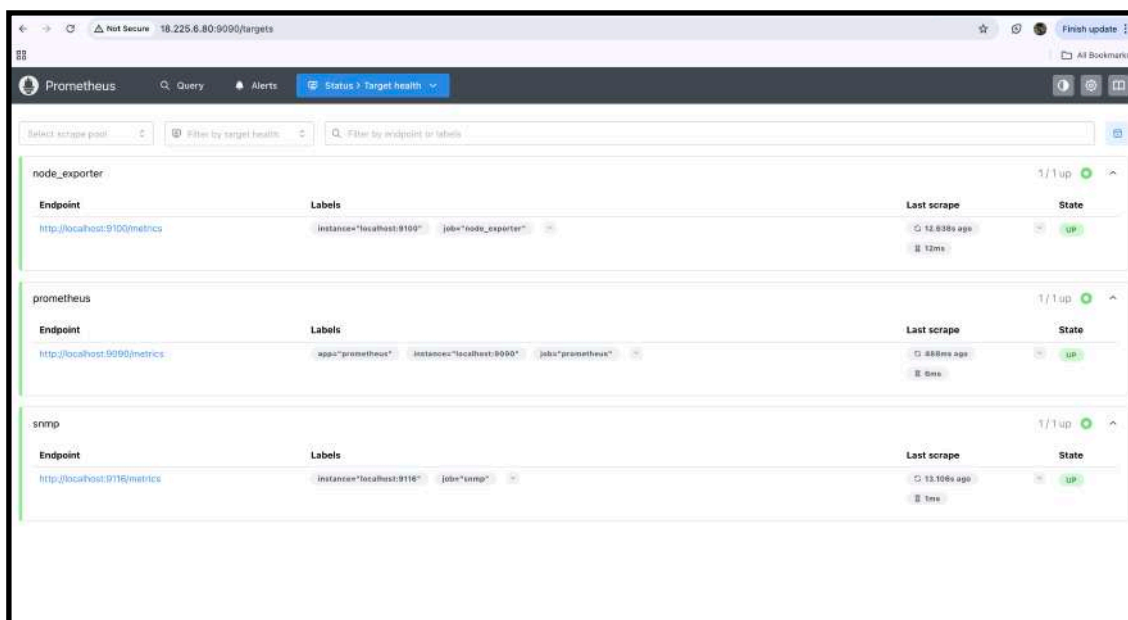
```
[root@ip-172-31-46-78 prometheus]# systemctl daemon-reload
[root@ip-172-31-46-78 prometheus]# systemctl stop prometheus
[root@ip-172-31-46-78 prometheus]# systemctl start prometheus
[root@ip-172-31-46-78 prometheus]#
```

**i-057d60391984ab236 (promethues-server)**

## Step 7 — Verify in Browser

Open:

<http://<EC2-PUBLIC-IP>:9090>



## Task 5 — Install Alertmanager and Configure Email Alert

## Step 1 — Go to /opt

```
cd /opt
```

## Step 2 — Download Alertmanager

```
wget https://github.com/prometheus/alertmanager/releases/  
download/v0.28.0/alertmanager-0.28.0.linux-amd64.tar.gz
```

## Step 3 — Extract the Package

```
tar -xvf alertmanager-0.28.0.linux-amd64.tar.gz
```

[illegible]

Rename folder:

```
mv alertmanager-0.28.0.linux-amd64 alertmanager
```



```
[root@ip-172-31-46-78 opt]# tar -xvf alertmanager-0.28.0.linux-amd64.tar.gz
alertmanager-0.28.0.linux-amd64/
alertmanager-0.28.0.linux-amd64/LICENSE
alertmanager-0.28.0.linux-amd64/alertmanager
alertmanager-0.28.0.linux-amd64/alertmanager.yml
alertmanager-0.28.0.linux-amd64/amtool
alertmanager-0.28.0.linux-amd64/NOTICE
[root@ip-172-31-46-78 opt]# mv alertmanager-0.28.0.linux-amd64 alertmanager
[root@ip-172-31-46-78 opt]# ls
alertmanager      aws               node_exporter-1.9.1.lin
alertmanager-0.28.0.linux-amd64.tar.gz  node_exporter    prometheus
[root@ip-172-31-46-78 opt]#
```

## Step 4 — Go Inside Directory

```
cd alertmanager
```

Check files:

```
ls
```

```
alertmanager-0.28.0.linux-amd64.tar.gz  node_exporter  prometheus
[root@ip-172-31-46-78 opt]# cd alertmanager/
[root@ip-172-31-46-78 alertmanager]# ls
LICENSE  NOTICE  alertmanager  alertmanager.yml  amtool
[root@ip-172-31-46-78 alertmanager]#
```

**i-057d60391984ab236 (promethues-server)**

## Step 5 — Configure Email Alert

Edit configuration file.

```
vi alertmanager.yml
```

Replace the configuration with:

global:

smtp\_smarthost: 'smtp.gmail.com:587'

smtp\_from: 'test@gmail.com'

smtp\_auth\_username: 'test@gmail.com'

smtp\_auth\_password: 'password'

route:

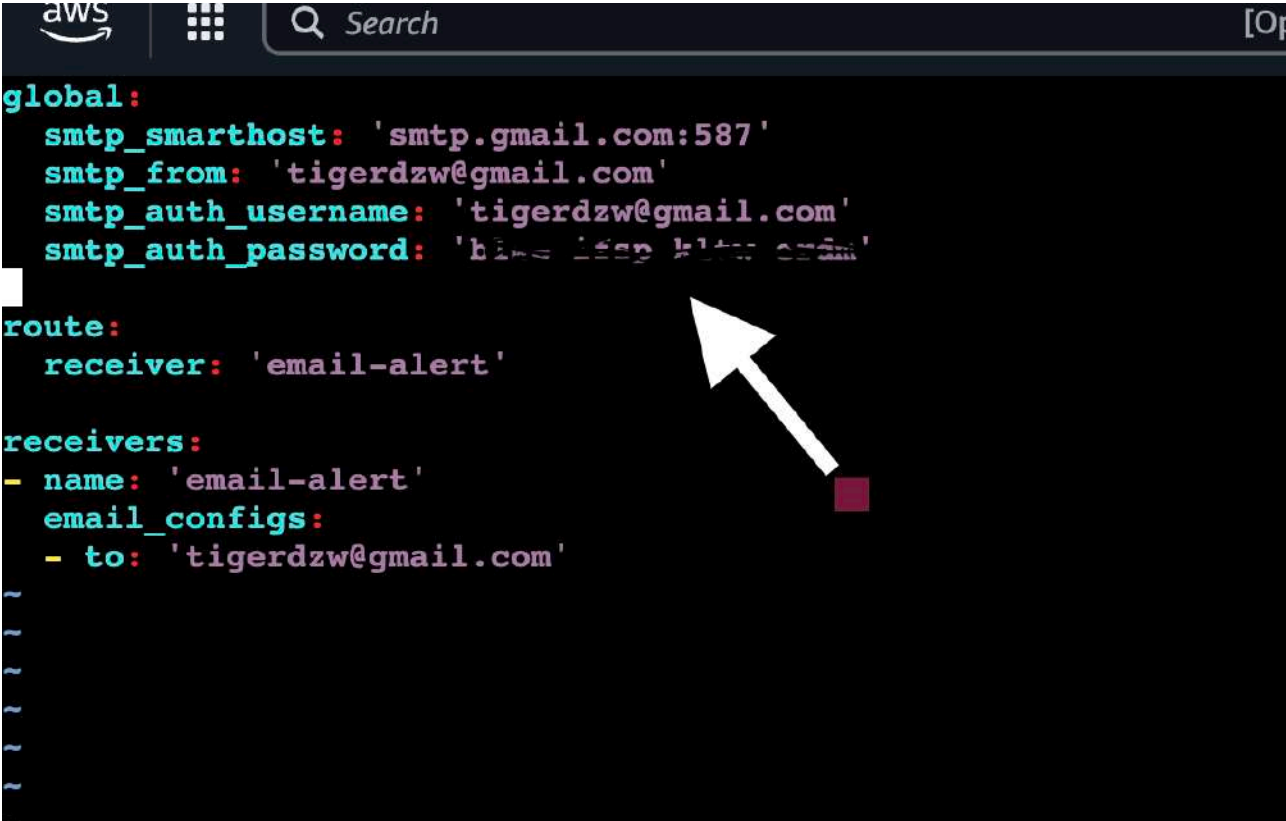
receiver: 'email-alert'

receivers:

- name: 'email-alert'

email\_configs:

- to: 'receiver@gmail.com'



```
aws | Search [Op
global:
  smtp_smarthost: 'smtp.gmail.com:587'
  smtp_from: 'tigerdzw@gmail.com'
  smtp_auth_username: 'tigerdzw@gmail.com'
  smtp_auth_password: 'h1ne 1fap kltu ordm'
route:
  receiver: 'email-alert'
receivers:
- name: 'email-alert'
  email_configs:
  - to: 'tigerdzw@gmail.com'
~
~
~
~
~
```

## Step 6 — Test Alertmanager

Run Alertmanager manually:

```
./alertmanager --config.file=alertmanager.yml
```

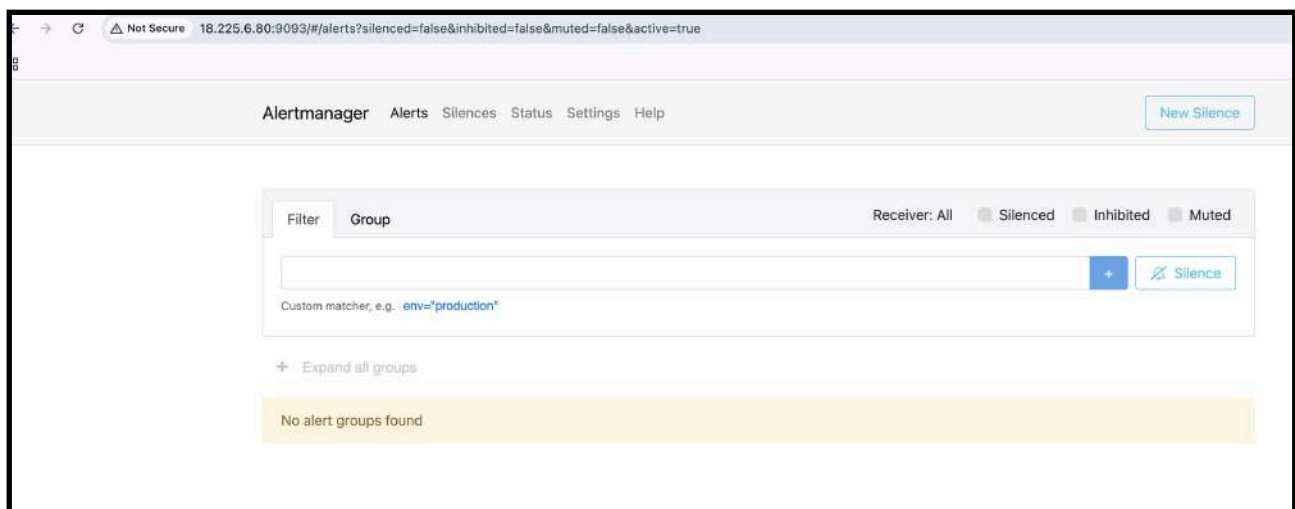
```
LICENSE NOTICE alertmanager alertmanager.yml amtool
[root@ip-172-31-46-78 alertmanager]# vi alertmanager.yml
[root@ip-172-31-46-78 alertmanager]# ./alertmanager --config.file=alertmanager.yml
time=2026-03-05T11:43:31.991Z level=INFO source=main.go:191 msg="Starting Alertmanager" version="(version=0.28.0, branch=
p)"
time=2026-03-05T11:43:31.992Z level=INFO source=main.go:192 msg="Build context" build_context="(go=go1.23.4, platform=lin
p)"
time=2026-03-05T11:43:31.992Z level=INFO source=cluster.go:185 msg="setting advertise address explicitly" component=clust
time=2026-03-05T11:43:31.995Z level=INFO source=cluster.go:674 msg="Waiting for gossip to settle..." component=cluster in
time=2026-03-05T11:43:32.026Z level=INFO source=coordinator.go:112 msg="Loading configuration file" component=configurati
time=2026-03-05T11:43:32.026Z level=INFO source=coordinator.go:125 msg="Completed loading of configuration file" componen
time=2026-03-05T11:43:32.029Z level=INFO source=tlc_config.go:347 msg="Listening on" address=[::]:9093
time=2026-03-05T11:43:32.029Z level=INFO source=tlc_config.go:350 msg="TLS is disabled." http2=false address=[::]:9093
time=2026-03-05T11:43:33.995Z level=INFO source=cluster.go:699 msg="gossip not settled" component=cluster polls=0 before=
time=2026-03-05T11:43:41.999Z level=INFO source=cluster.go:691 msg="gossip settled; proceeding" component=cluster elapsed
i-057d60391984ab236 (promethues-server)
```

## Step 7 — Verify Alertmanager UI

Open browser:

<http://<EC2-PUBLIC-IP>:9093>

If the **Alertmanager dashboard** opens, installation is correct.

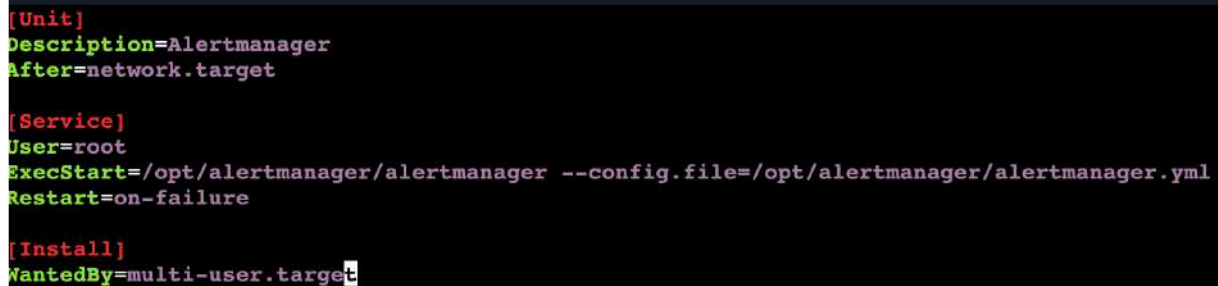


## Step 8 — Create Alertmanager Service

Create service file:

```
vi /etc/systemd/system/alertmanager.service
```

Add this configuration:

A screenshot of a terminal window with a dark background. The terminal shows the configuration for the alertmanager.service file. The configuration is divided into three sections: [Unit], [Service], and [Install]. The [Unit] section contains Description=Alertmanager and After=network.target. The [Service] section contains User=root, ExecStart=/opt/alertmanager/alertmanager --config.file=/opt/alertmanager/alertmanager.yml, and Restart=on-failure. The [Install] section contains WantedBy=multi-user.target. The cursor is at the end of the line WantedBy=multi-user.target.

```
[Unit]
Description=Alertmanager
After=network.target

[Service]
User=root
ExecStart=/opt/alertmanager/alertmanager --config.file=/opt/alertmanager/alertmanager.yml
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

## Step 9 — Reload systemd

```
systemctl daemon-reload
```

## Step 10 — Start Alertmanager

```
systemctl start alertmanager
```

Enable it on boot:

```
systemctl enable alertmanager
```

## Step 11 — Verify Service

```
systemctl status alertmanager
```

```

time=2026-03-05T11:46:52.657Z level=INFO source=main.go:584 msg="Received SIGTERM, exiting gracefully..."
[root@ip-172-31-46-78 alertmanager]# vi /etc/systemd/system/alertmanager.service
[root@ip-172-31-46-78 alertmanager]# systemctl daemon-reload
[root@ip-172-31-46-78 alertmanager]# systemctl start alertmanager
[root@ip-172-31-46-78 alertmanager]# systemctl enable alertmanager
Created symlink /etc/systemd/system/multi-user.target.wants/alertmanager.service → /etc/systemd/system/alertmanager.service.
[root@ip-172-31-46-78 alertmanager]# systemctl status alertmanager
● alertmanager.service - Alertmanager
   Loaded: loaded (/etc/systemd/system/alertmanager.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 11:48:18 UTC; 8s ago
     Main PID: 5892 (alertmanager)
        Tasks: 7 (limit: 1067)
       Memory: 13.3M
          CPU: 77ms
      CGroup: /system.slice/alertmanager.service
              └─5892 /opt/alertmanager/alertmanager --config.file=/opt/alertmanager/alertmanager.yml

Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal systemd[1]: Started alertmanager.service - Alertmanager.
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.808Z level=INFO
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.809Z level=INFO
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.812Z level=INFO
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.843Z level=INFO
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.843Z level=INFO
Mar 05 11:48:18 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:18.846Z level=INFO
Mar 05 11:48:20 ip-172-31-46-78.us-east-2.compute.internal alertmanager[5892]: time=2026-03-05T11:48:20.813Z level=INFO
lines 1-20/20 (END)

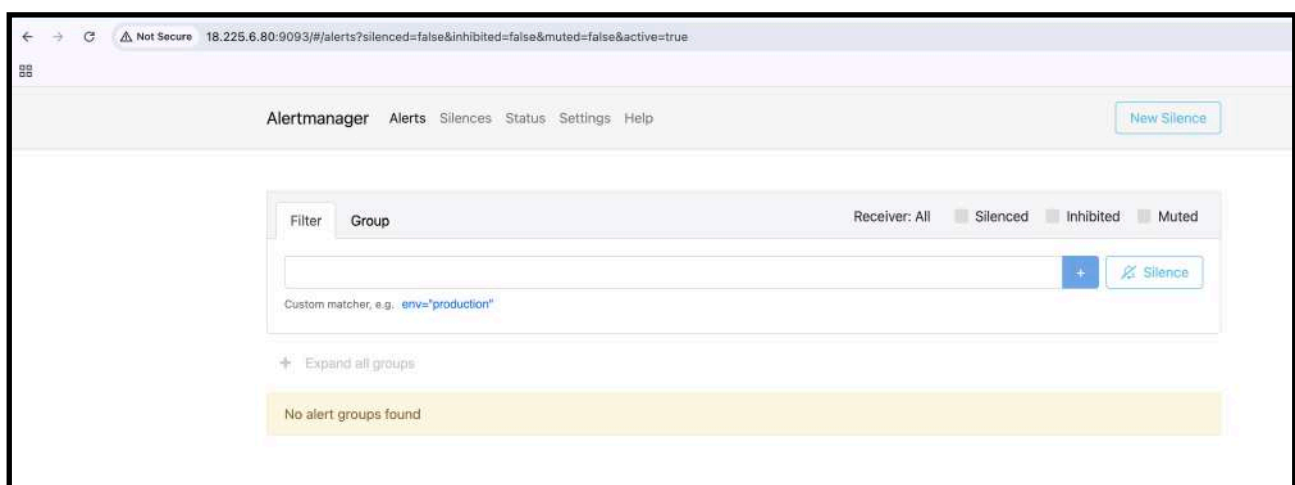
```

## Step 12 — Final Verification

Open:

<http://<EC2-PUBLIC-IP>:9093>

You should see the **Alertmanager** dashboard.



# Task 5 Completed

Alertmanager is now installed and configured to send **email alerts**.

## Task 6 — Install Grafana and Add Prometheus as Data Source

### Step 1 — Install Grafana

Since your server looks like **Amazon Linux**, run:

```
sudo yum install -y https://dl.grafana.com/oss/release/
grafana-10.4.3-1.x86_64.rpm
```

```
[root@ip-172-31-46-78 alertmanager]# sudo yum install -y https://dl.grafana.com/oss/release/grafana-10.4.3-1.x86_64.rpm
Last metadata expiration check: 2:04:00 ago on Thu Mar  5 09:52:24 2026.
grafana-10.4.3-1.x86_64.rpm
Dependencies resolved.
=====
Package                                Architecture      Version            Repository
=====
Installing:
grafana                                x86_64            10.4.3-1           @commandlin
Installing dependencies:
cairo                                  x86_64            1.18.0-4.amzn2023.0.3  amazonlinu
fontconfig                             x86_64            2.13.94-2.amzn2023.0.2  amazonlinu
fonts-filesystem                       noarch            1:2.0.5-12.amzn2023.0.2  amazonlinu
freetype                               x86_64            2.13.2-5.amzn2023.0.1  amazonlinu
google-noto-fonts-common              noarch            20240401-1.amzn2023.0.2  amazonlinu
google-noto-sans-vf-fonts            noarch            20240401-1.amzn2023.0.2  amazonlinu
graphite2                              x86_64            1.3.14-7.amzn2023.0.2  amazonlinu
harfbuzz                               x86_64            7.0.0-2.amzn2023.0.2  amazonlinu
langpacks-core-font-en               noarch            3.0-21.amzn2023.0.4  amazonlinu
libX11                                 x86_64            1.8.10-2.amzn2023.0.1  amazonlinu
libX11-common                        noarch            1.8.10-2.amzn2023.0.1  amazonlinu
libXau                                x86_64            1.0.11-6.amzn2023.0.1  amazonlinu
libXext                               x86_64            1.3.6-1.amzn2023.0.1  amazonlinu
libXrender                           x86_64            0.9.11-6.amzn2023.0.1  amazonlinu
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2  amazonlinu
libpng                                 x86_64            2:1.6.37-10.amzn2023.0.10  amazonlinu
libxcb                                 x86_64            1.17.0-1.amzn2023.0.1  amazonlinu
pixman                                x86_64            0.43.4-1.amzn2023.0.4  amazonlinu
xml-common                            noarch            0.6.3-56.amzn2023.0.2  amazonlinu
=====
Transaction Summary
-----
Install 20 Packages
```

### Step 2 — Start Grafana

```
sudo systemctl start grafana-server
```



Enable it on boot:

```
sudo systemctl enable grafana-server
```

## Step 3 — Verify Grafana Service

Run:

```
sudo systemctl status grafana-server
```

```
Complete!
[root@ip-172-31-46-78 alertmanager]# sudo systemctl start grafana-server
[root@ip-172-31-46-78 alertmanager]# sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
[root@ip-172-31-46-78 alertmanager]# sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-05 11:57:34 UTC; 7s ago
     Docs: http://docs.grafana.org
  Main PID: 29739 (grafana)
    Tasks: 8 (limit: 1067)
   Memory: 168.1M
      CPU: 2.308s
   CGroup: /system.slice/grafana-server.service
           └─29739 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/var/run/grafana/grafana.pid

Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=renderer.manager t=2026-03-05T11:57:35.00
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=provisioning.dashboard t=2026-03-05T11:57
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=provisioning.dashboard t=2026-03-05T11:57
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=ngalert.multiorg.alertmanager t=2026-03-0
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=ngalert.scheduler t=2026-03-05T11:57:35.0
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=ticker t=2026-03-05T11:57:35.019837544Z 1
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=plugins.update.checker t=2026-03-05T11:57
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=grafana.update.checker t=2026-03-05T11:57
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=grafana-apiserver t=2026-03-05T11:57:35.5
Mar 05 11:57:35 ip-172-31-46-78.us-east-2.compute.internal grafana[29739]: logger=grafana-apiserver t=2026-03-05T11:57:35.5
lines 1-21/21 (END)
```

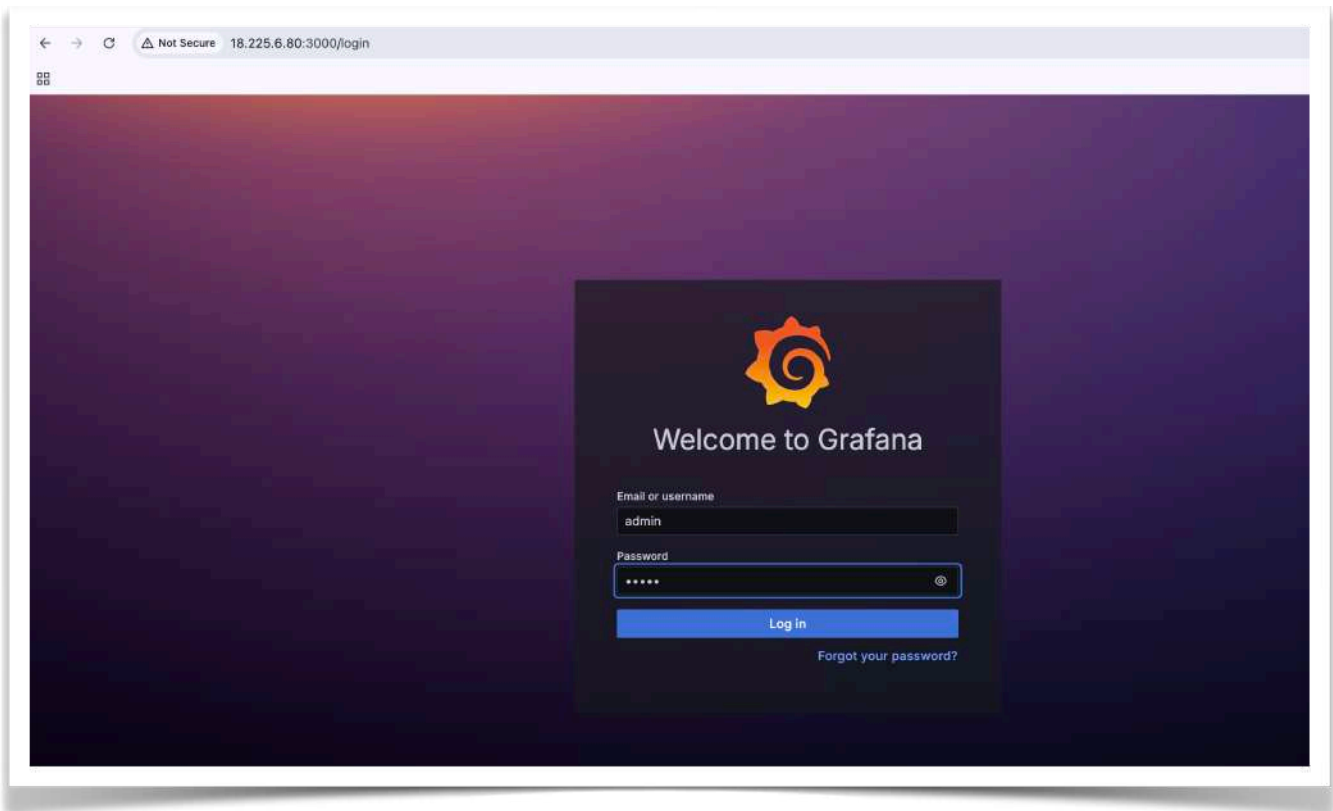
## Step 4 — Open Grafana UI

Open browser:

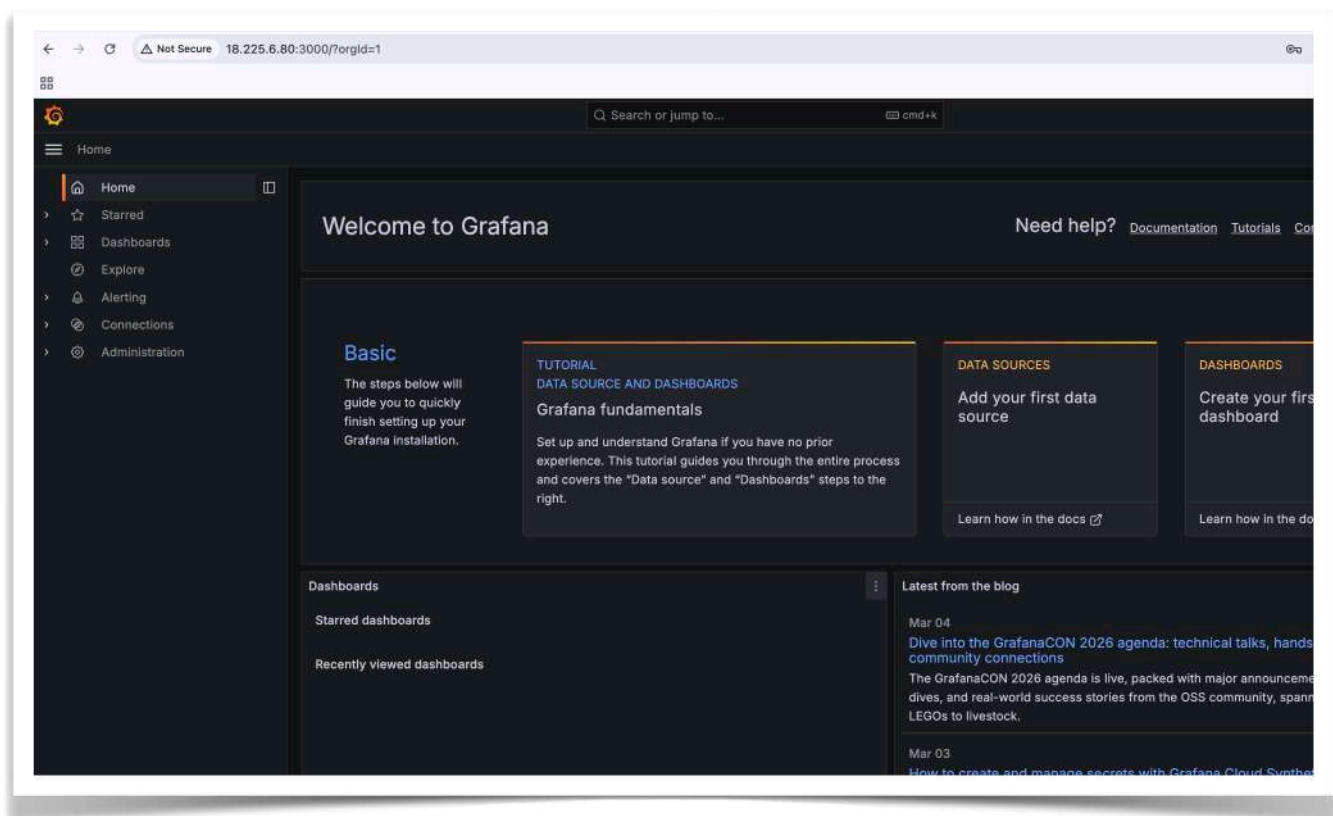
```
http://<EC2-PUBLIC-IP>:3000
```

Default login:

Username: admin  
Password: admin



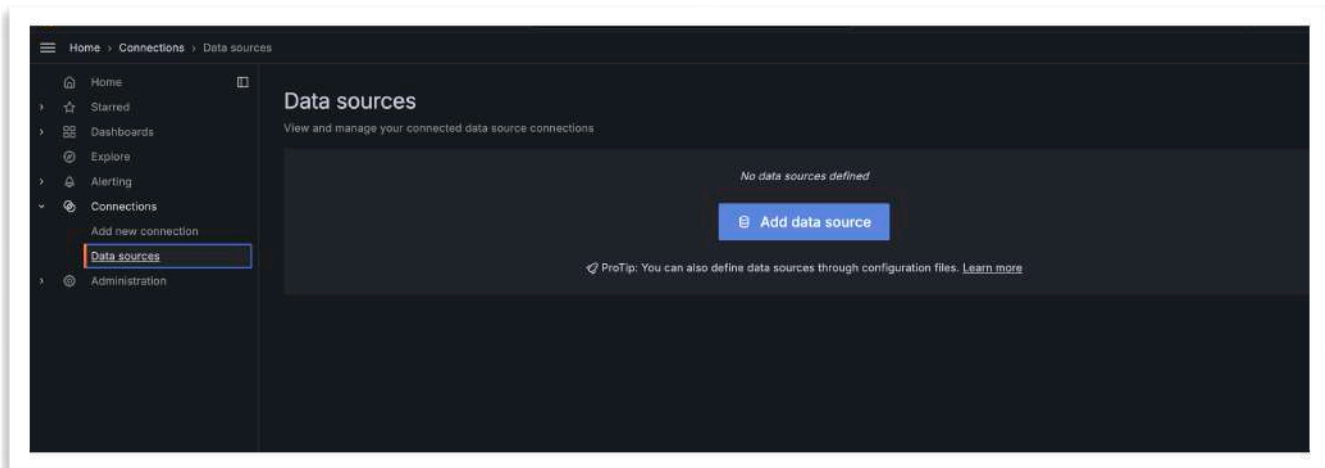
Grafana will ask you to **set a new password**.



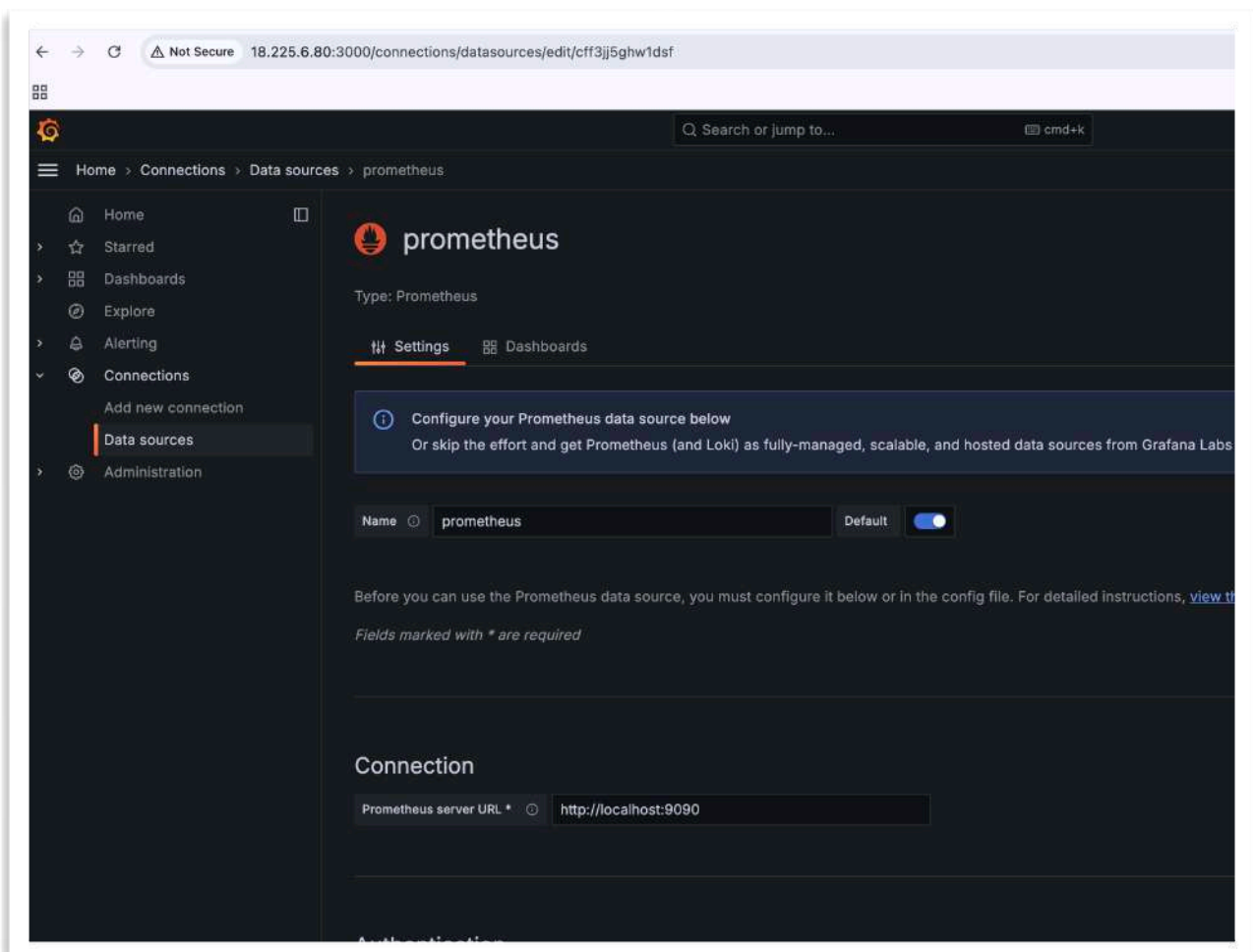
## Step 5 — Add Prometheus Data Source

Inside Grafana:

- 1 Click **Connections**
- 2 Click **Data Sources**
- 3 Click **Add data source**
- 4 Select **Prometheus**



—> since we using same server use localhost:9090

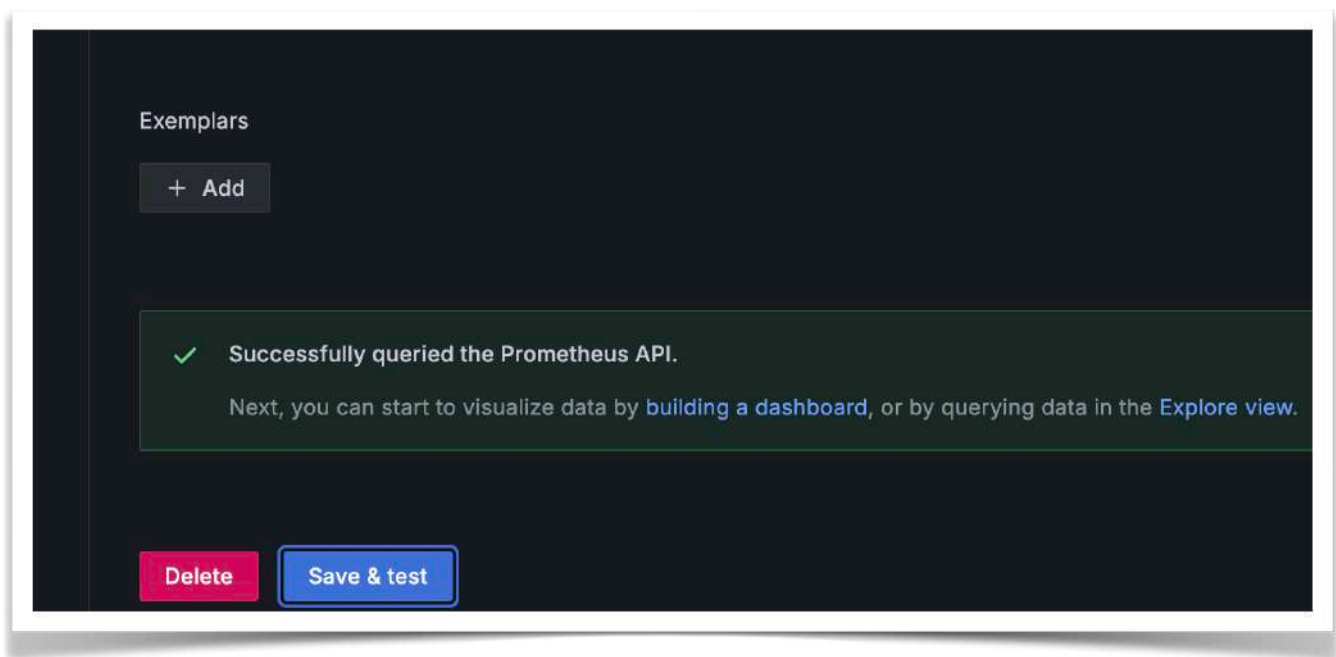


Then click:

Save & Test

You should see:

Data source is working



## Step 7 — Verify Metrics

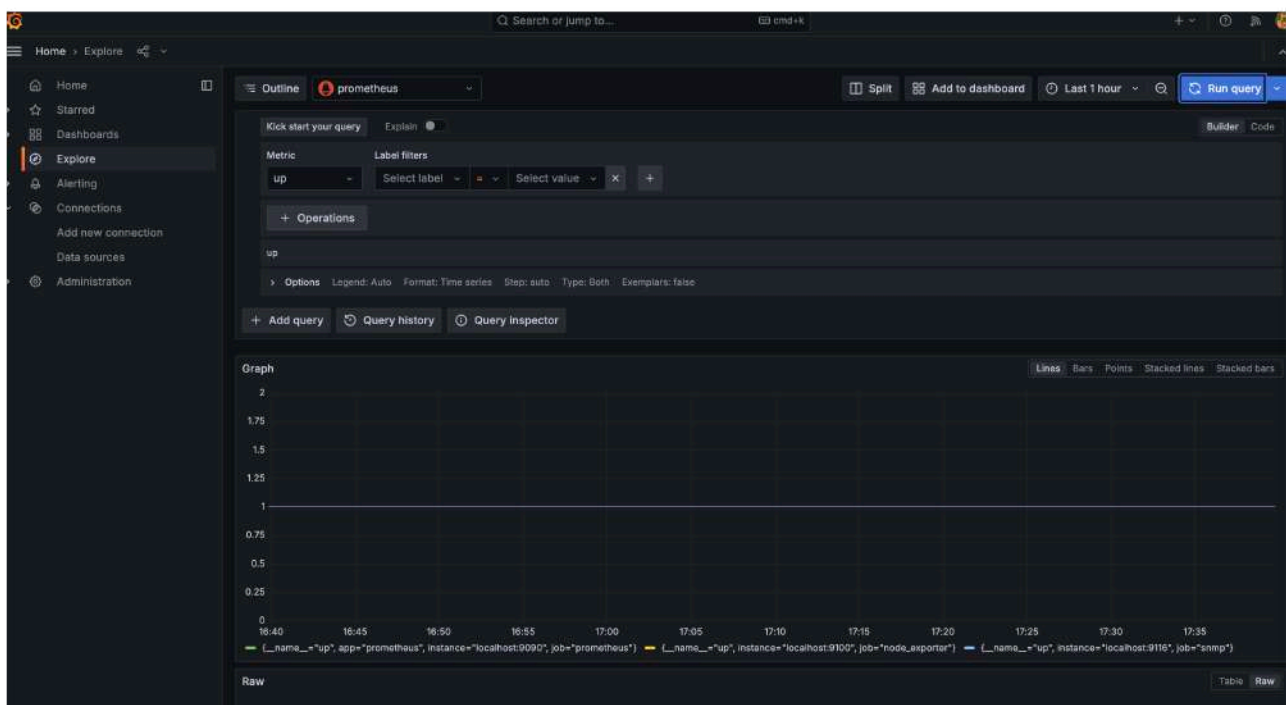
Go to:

Explore

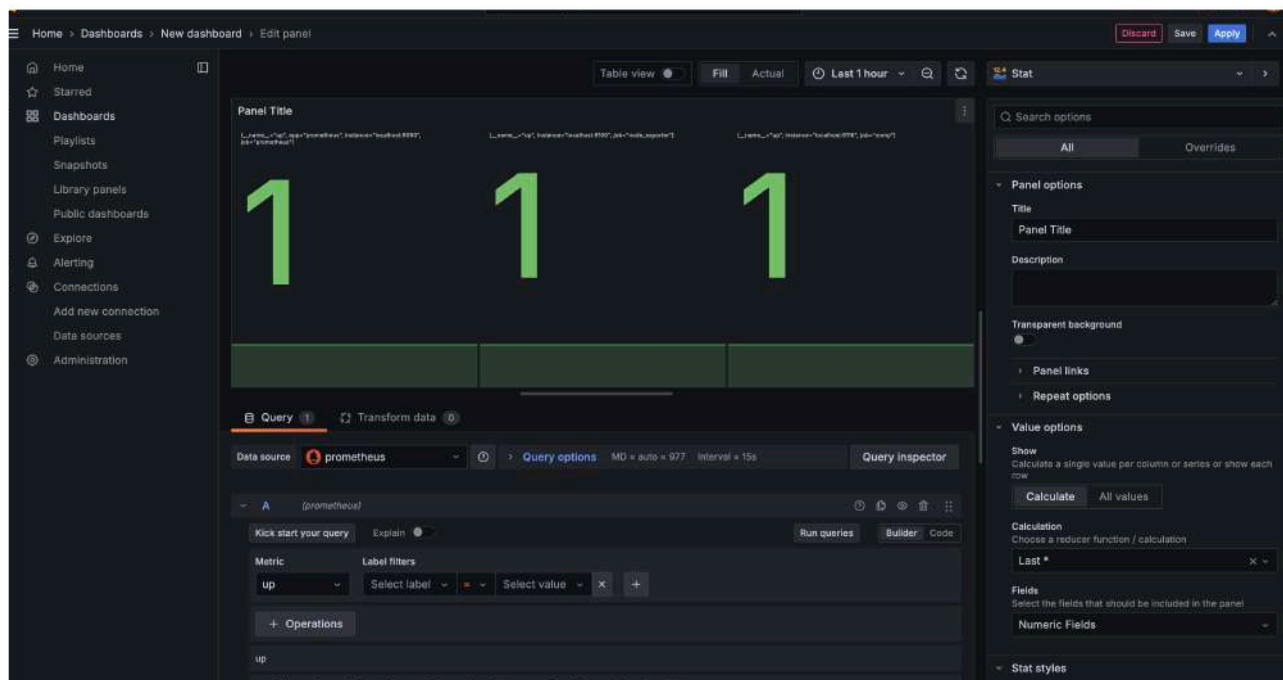
Select **Prometheus** and run query:

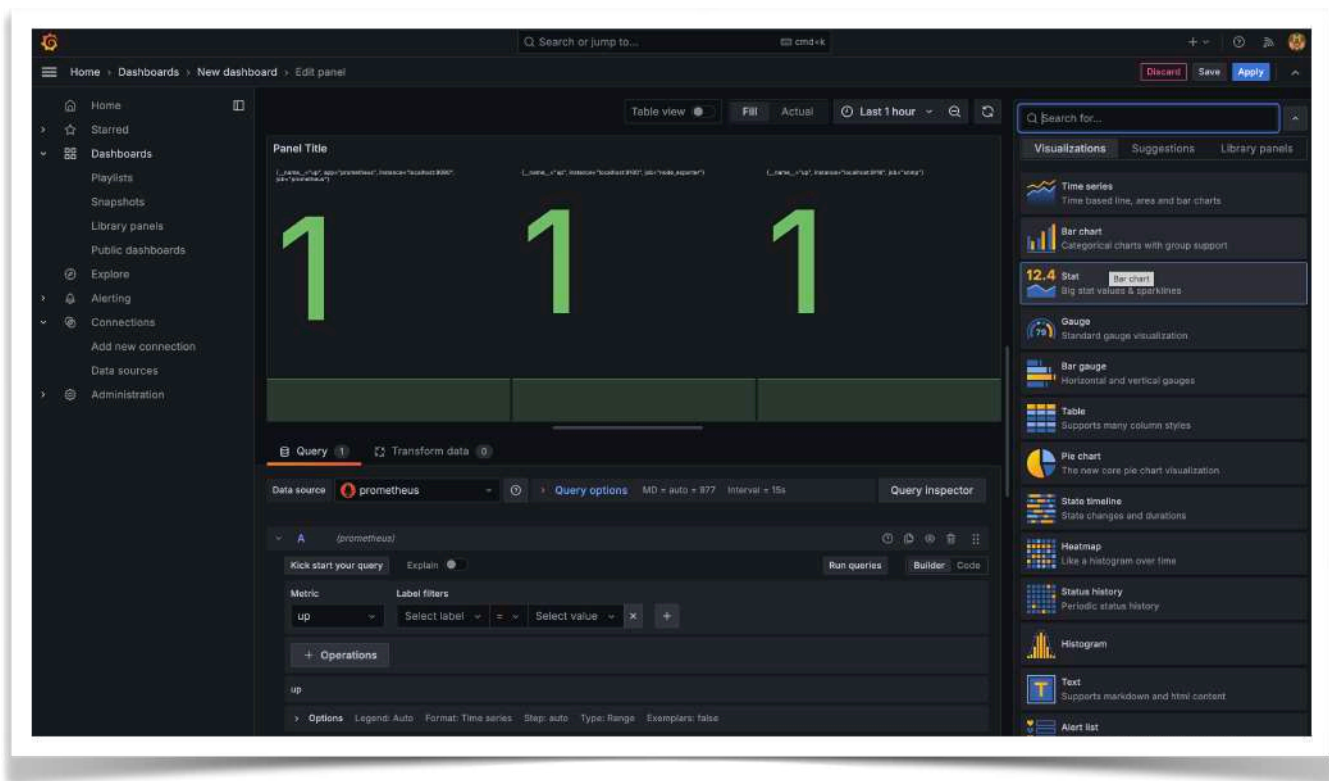
up

You should see metrics.



—> we are able to check the query from dashboard visualisation





## Task 7 — Configure Prometheus for Service Discovery of Dynamic Targets

### Step 1 — Launch a New EC2 Instance (Target Node)

Create a second EC2 instance.

Example:

- **Name:** prometheus-node-1
- **AMI:** Amazon Linux
- **Instance Type:** t3.micro

This instance will be the **target machine** monitored by Prometheus.

| Name            | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS           | Public |
|-----------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|---------------------------|--------|
| prometheus-s... | i-057d60391984ab236 | Running        | t3.micro      | 3/3 checks passed | View alarms + | us-east-2c        | ec2-18-225-6-80.us-eas... | 18.225 |
| promet-node-1   | i-0db05753111f43820 | Running        | t3.micro      | Initializing      | View alarms + | us-east-2c        | ec2-18-118-212-33.us-...  | 18.118 |



## Step 2 — Install Node Exporter on Target Instance

## SSH into the new EC2 instance.

```
cd /opt
```

Download node exporter:

```
wget https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
```

Extract it:

```
tar -xvf node_exporter-1.9.1.linux-amd64.tar.gz
mv node exporter-1.9.1.linux-amd64 node exporter
```

Go inside:

```
cd node_exporter
```

```
[root@ip-172-31-38-56 opt]# wget https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
--2026-03-05 12:22:11-- https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.113.4
Connecting to github.com (github.com)[140.82.113.4]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/952405f2/dc8ec09c-2975-42a2-9591-57dd1ffff7b7?sp=r&sv=2018-11-09&s=8A46de&rscd=attachment%3B&filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rscst=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sk=b4deskt=2026-03-05T12%3A14%3A16Z&ske=2026-03-05T13%3A14%3A46Z&skv=2018-11-09&sig=mpsRG3kxANHU6EhI18WPUXW7iOwACegf8%2FhXt2B7XHTYg0%3D&jwt=eYJ0eXA1ODIjJnAXRodWlUy29tIiwiaXVzIjoicmVsZWZrSzhlc3NldHMud210aHViZDNlcmVbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjcNTeZSMwibmJmIjojNzcyczNzcyNmXLCJwYXRoIjoia2NmVmVud2luZG93cy5uZQkiQ.GqKyf9FUtkfgXNKXuN2H22zi96Ti_PxZh4UHpvqD7zc&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.9.1.linux-application%2Foctet-stream [following]
--2026-03-05 12:22:11-- https://release-assets.githubusercontent.com/github-production-release-asset/952405f2/dc8ec09c-2975-42a2-9591-57dd1ffff7b7?sp=r&-03-05T13%3A14%3A46Z&rscd=attachment%3B&filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rscst=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-9b12b-9515b896b4deskt=2026-03-05T12%3A14%3A16Z&ske=2026-03-05T13%3A14%3A46Z&skv=2018-11-09&sig=mpsRG3kxANHU6EhI18WPUXW7iOwACegf8%2FhXt2B7XHTYg0%3D1ni39.eYJpc3MiOiJnAXRodWlUy29tIiwiaXVzIjoicmVsZWZrSzhlc3NldHMud210aHViZDNlcmVbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjcNTeZSMwibmJmIjojNzcyczNzcyNmXLCJwYXRoIjoia2NmVmVud2luZG93cy5uZQkiQ.GqKyf9FUtkfgXNKXuN2H22zi96Ti_PxZh4UHpvqD7zc&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.9.1.linux-application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11582410 (11M) [application/octet-stream]
Saving to: 'node_exporter-1.9.1.linux-amd64.tar.gz'

node_exporter-1.9.1.linux-amd64.tar.gz      100%[=====
2026-03-05 12:22:11 (235 MB/s) - 'node_exporter-1.9.1.linux-amd64.tar.gz' saved [11582410/11582410]
```

```
[root@ip-172-31-38-56 opt]# tar -xvf node_exporter-1.9.1.linux-amd64.tar.gz
mv node_exporter-1.9.1.linux-amd64 node_exporter
node_exporter-1.9.1.linux-amd64/
node_exporter-1.9.1.linux-amd64/LICENSE
node_exporter-1.9.1.linux-amd64/NOTICE
node_exporter-1.9.1.linux-amd64/node_exporter
[root@ip-172-31-38-56 opt]# ls
aws node_exporter node_exporter-1.9.1.linux-amd64.tar.gz
[root@ip-172-31-38-56 opt]# cd node_exporter/
[root@ip-172-31-38-56 node_exporter]# ls
LICENSE NOTICE node_exporter
[root@ip-172-31-38-56 node_exporter]#
```

Start exporter:

```
./node_exporter
```

Verify:

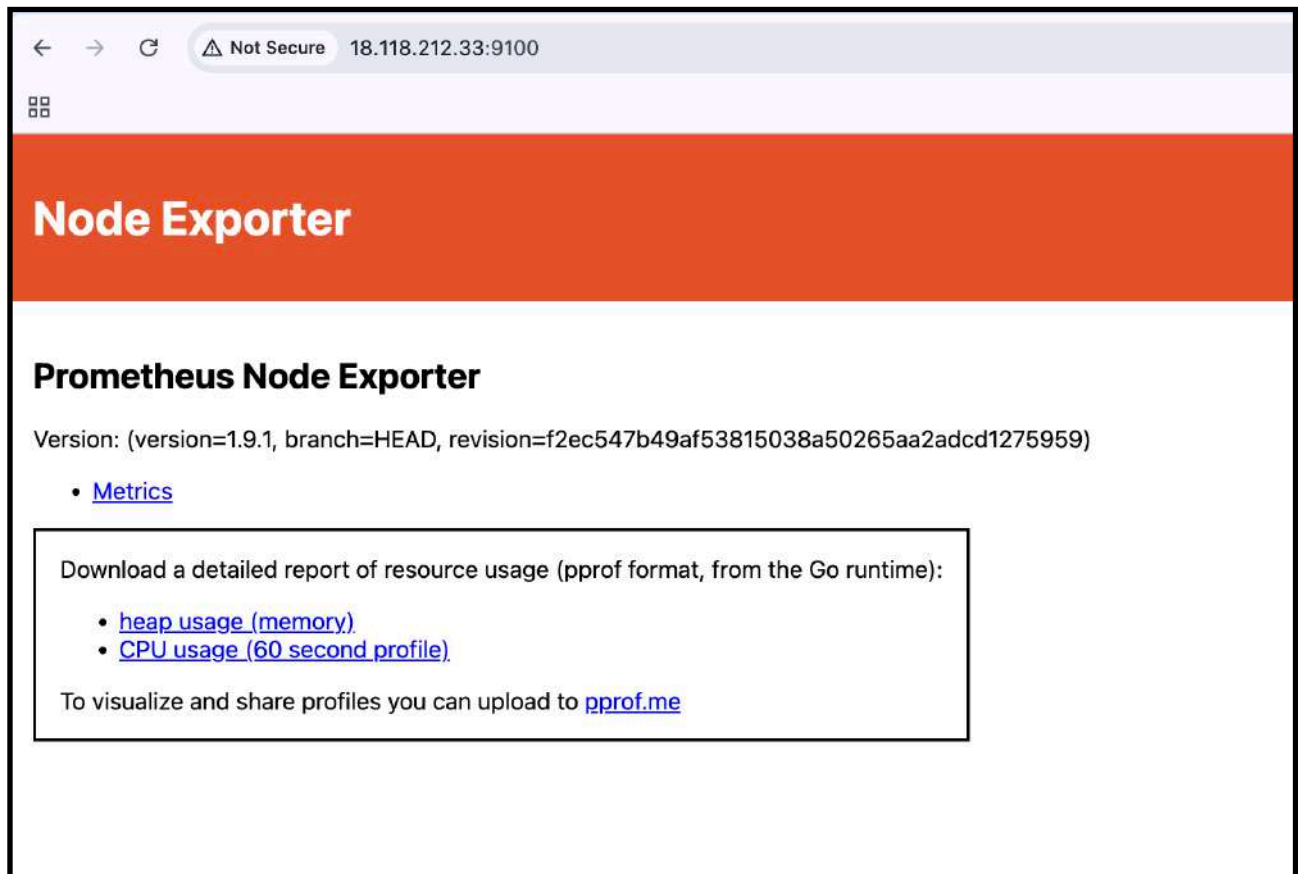
[http://TARGET\\_PUBLIC\\_IP:9100/metrics](http://TARGET_PUBLIC_IP:9100/metrics)

If metrics appear → Node Exporter is working.

```

LICENSE NOTICE node_exporter
[root@ip-172-31-38-56 node_exporter]# ./node_exporter
time=2026-03-05T12:24:06.189Z level=INFO source=node_exporter.go:216 msg="Starting node_exporter" v
time=2026-03-05T12:24:06.189Z level=INFO source=node_exporter.go:217 msg="Build context" build_cont
tags=unknown)"
time=2026-03-05T12:24:06.190Z level=WARN source=node_exporter.go:219 msg="Node Exporter is running
."
time=2026-03-05T12:24:06.190Z level=INFO source=filesystem_common.go:265 msg="Parsed flag --collect
/./+|sys|var/lib/docker/./+|var/lib/containers/storage/./+)($/|/)
time=2026-03-05T12:24:06.190Z level=INFO source=filesystem_common.go:294 msg="Parsed flag --collect
2?|configfs|debugfs|devpts|devtmpfs|fusectl|hugetlbfs|iso9660|mqueue|nsfs|overlay|proc|procfs|pstor
time=2026-03-05T12:24:06.191Z level=INFO source=diskstats_common.go:110 msg="Parsed flag --collecto
e\d+n\d+p)\d+$
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:135 msg="Enabled collectors"
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=arp
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=bcache
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=bonding
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=btrfs
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=conntrack
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=cpu
time=2026-03-05T12:24:06.192Z level=INFO source=node_exporter.go:141 msg=cputime

```



You can see:

| Key  | Value         |
|------|---------------|
| Name | promet-node-1 |

That means the **EC2 instance already has the tag.**



Prometheus will use this tag to discover the instance.

# Step 4 — Create IAM Role for Prometheus Server

Go to:

AWS Console → IAM → Roles → Create role

Select:

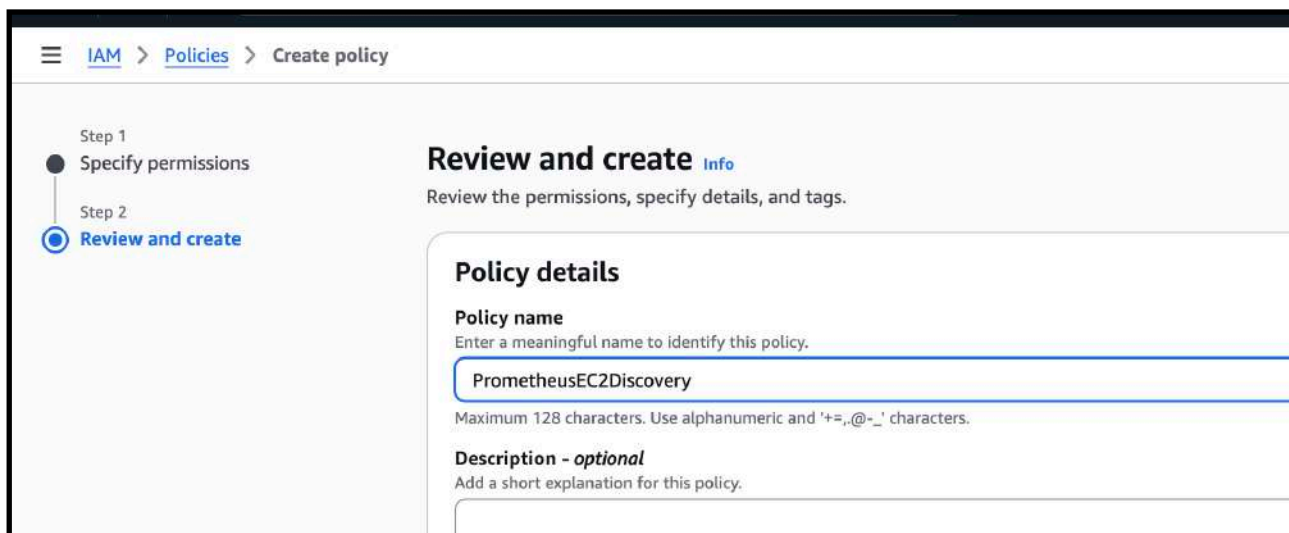
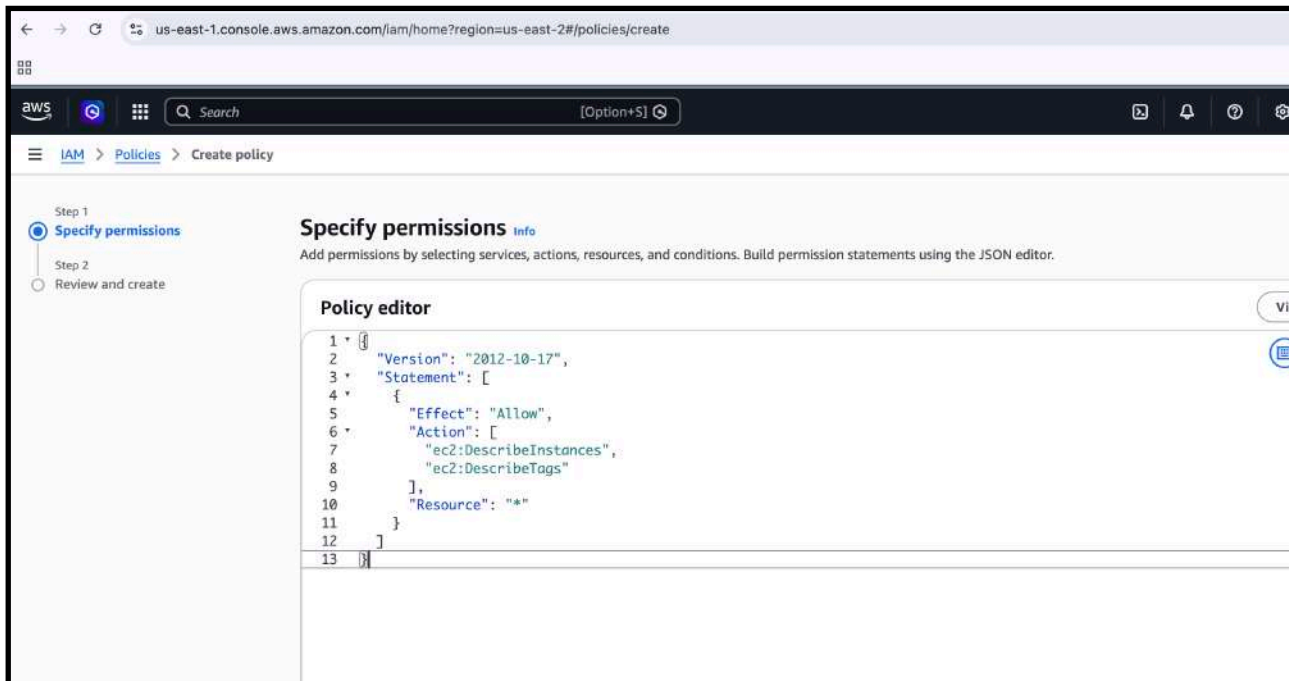
AWS Service → EC2

Create policy with this JSON:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "ec2:DescribeTags"
      ],
      "Resource": "*"
    }
  ]
}
```

Name policy:

PrometheusEC2Discovery



—> now create a role with this policy

Create role:

IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

**Name, review, and create**

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Prometheus-EC2-Discovery-Role

Maximum 64 characters. Use alphanumeric and '+\*,@-.' characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: \_+\*, @-/\[\]!\$%^&'`~="

Step 1: Select trusted entities

Trust policy

```
1 - [{
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    }
15  ]
16 }
```

## Prometheus-EC2-Discovery-Role

## Step 5 — Attach Role to Prometheus Server

Go to:

## EC2 → Instances

## Select Prometheus server instance

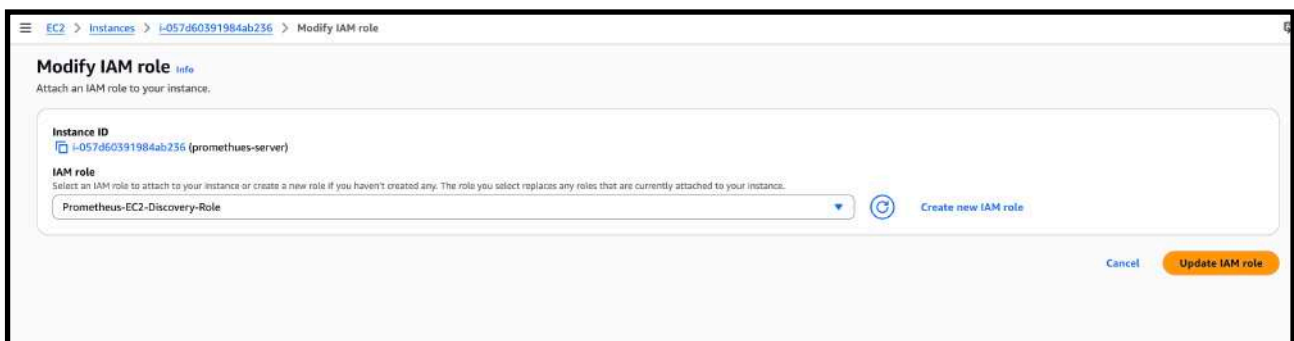
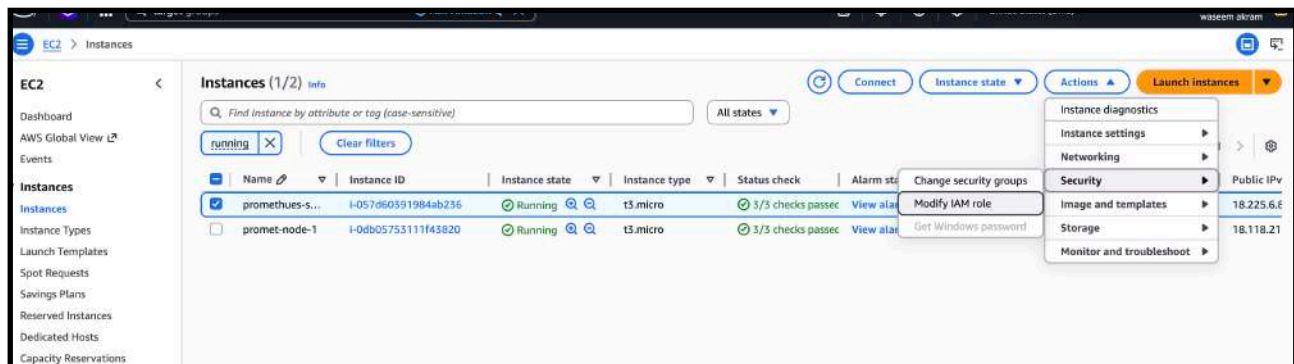
Click:



Actions → Security → Modify IAM role

Attach:

Prometheus-EC2-Discovery-Role



## Step 6 — Update Prometheus Configuration

SSH into **Prometheus server**.

Edit:

```
vi /opt/prometheus/prometheus.yml
```

Add this inside `scrape_configs`:

```
- job_name: 'ec2-node-exporter'
  ec2_sd_configs:
    - region: us-east-2
      port: 9100
  relabel_configs:
    - source_labels: [__meta_ec2_tag_Name]
      regex: prometheus.*
      action: keep
```

```
# A scrape configuration containing exactly one endpoint to scrape.
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label `label_name=<label_value>` to any timeseries scraped from this config.
        labels:
          app: "prometheus"

  - job_name: "node_exporter"
    static_configs:
      - targets: ["localhost:9100"]
  - job_name: "snmp"
    static_configs:
      - targets: ["localhost:9116"]

  - job_name: 'ec2-node-exporter'
    ec2_sd_configs:
      - region: us-east-2
        port: 9100
    relabel_configs:
      - source_labels: [__meta_ec2_tag_Name]
        regex: prometheus.*
        action: keep
-- INSERT --
```

```
i-057d60391984ab236 (prometheus-server)
```

## Step 7 — Restart Prometheus

```
systemctl restart prometheus
```

```
[root@ip-172-31-46-78 alertmanager]# vi /opt/prometheus/prometheus.yml
[root@ip-172-31-46-78 alertmanager]# vi /opt/prometheus/prometheus.yml
[root@ip-172-31-46-78 alertmanager]# systemctl restart prometheus
[root@ip-172-31-46-78 alertmanager]#
```

## Step 8 — Verify Service Discovery

Open Prometheus UI:

`http://PROMETHEUS_SERVER_IP:9090`

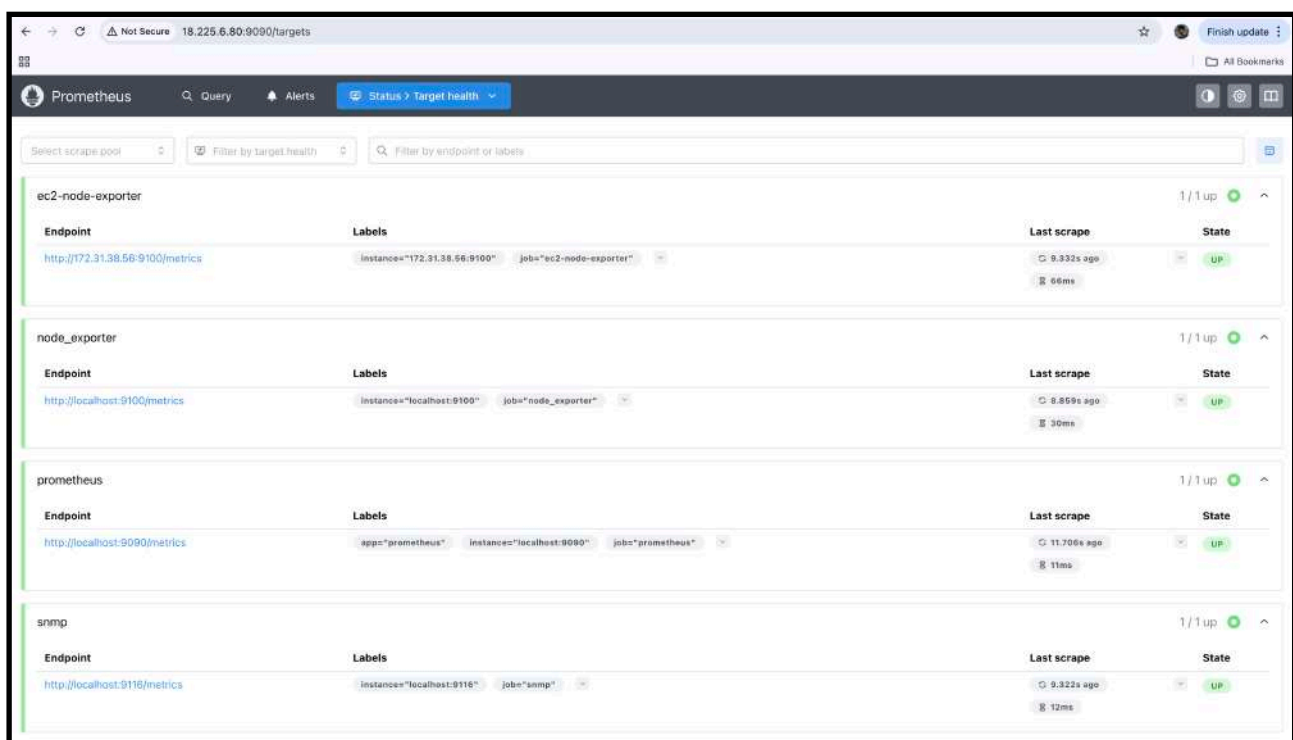
Go to:

Status → Targets

You should see:

`promet-node-1:9100`

Prometheus discovered the instance automatically.



# Task 7 Completed

Prometheus is now **discovering EC2 instances dynamically**.

## Task 8 — Create alerting rules for monitoring disk space.

### Create Alerting Rules for Disk Space

#### Step 1 — Go to Prometheus Directory

SSH into the **Prometheus server** and run:

```
cd /opt/prometheus
```

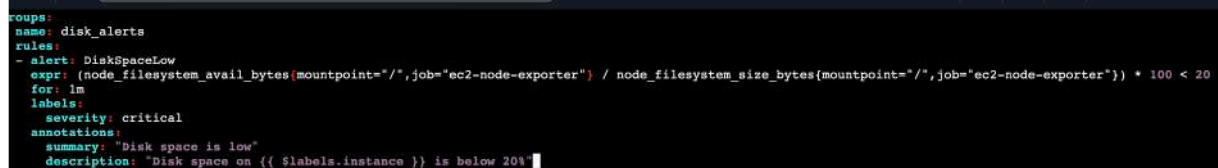
#### Step 2 — Create Alert Rule File

Create a new file:

```
vi disk_alert.yml
```

#### Step 3 — Add Disk Alert Rule

Paste this inside the file:



```
groups:
  name: disk_alerts
  rules:
    - alert: DiskSpaceLow
      expr: (node_filesystem_avail_bytes{mountpoint="/",job="ec2-node-exporter"} / node_filesystem_size_bytes{mountpoint="/",job="ec2-node-exporter"}) * 100 < 20
      for: 1m
      labels:
        severity: critical
      annotations:
        summary: "Disk space is low"
        description: "Disk space on {{ $labels.instance }} is below 20%"
```

This means:

Save the file:

| Condition | Meaning                                   |
|-----------|-------------------------------------------|
| < 20      | Alert triggers when disk free space < 20% |
| for: 1m   | Condition must remain for 1 minute        |

ESC

:wq

## Step 4 — Add Rule File in Prometheus Config

Edit the Prometheus configuration:

```
vi /opt/prometheus/prometheus.yml
```

```
# - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'eval_interval'
rule_files:
  - "disk_alert.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped here.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label `label_name=<label_value>` to any
        labels:
          app: "prometheus"

  - job_name: "node_exporter"
    static_configs:
      - targets: ["localhost:9100"]
  - job_name: "snmp"
    static_configs:
      - targets: ["localhost:9116"]

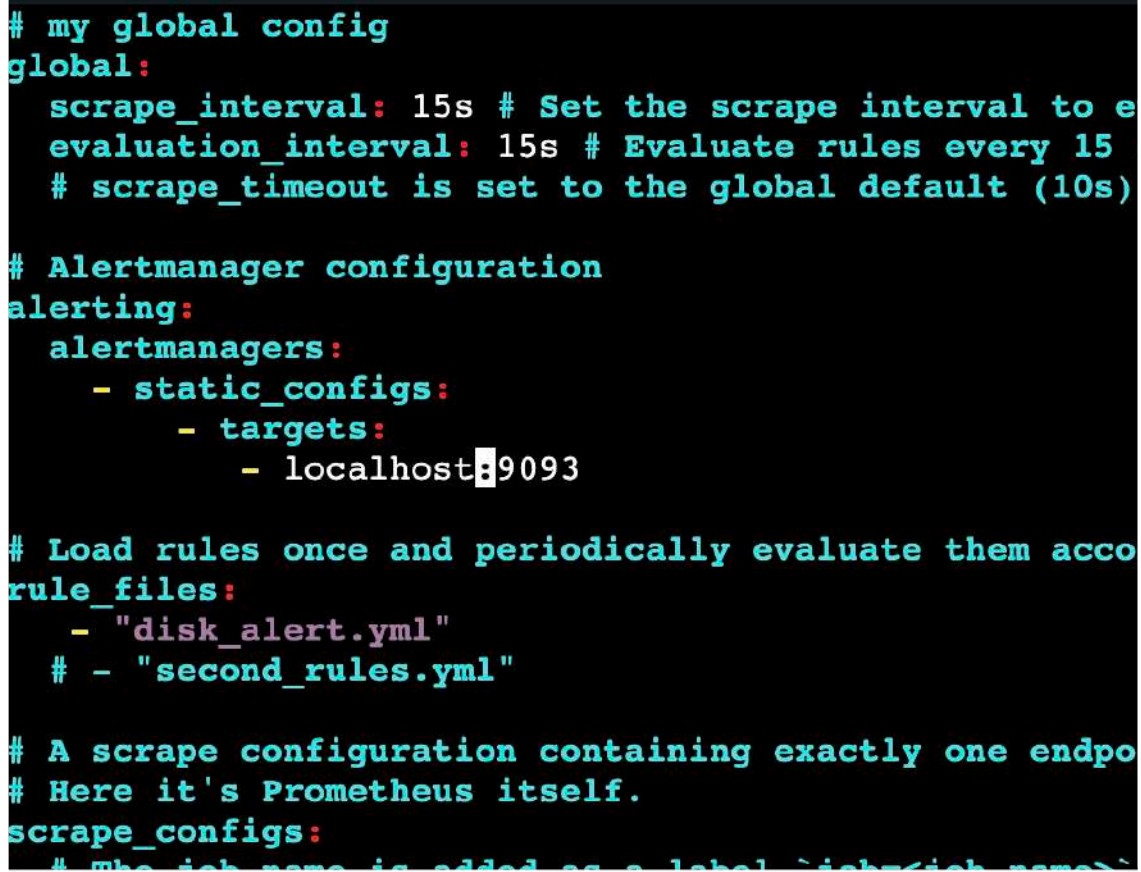
  - job_name: 'ec2-node-exporter'
    ec2_sd_configs:
      - region: us-east-2
        port: 9100
    relabel_configs:
      - source_labels: [__meta_ec2_tag_Name]
        regex: prometheus.*
        action: keep

-- INSERT --
```

## Step 4— Connect Prometheus to Alertmanager

In the same file (`prometheus.yml`) ensure this exists:

```
alerting:
  alertmanagers:
    - static_configs:
      - targets:
        - localhost:9093
```



```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds by default
  evaluation_interval: 15s # Evaluate rules every 15 seconds by default
  # scrape_timeout is set to the global default (10s)

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
      - targets:
        - localhost:9093

# Load rules once and periodically evaluate them according to the global
rule_files:
  - "disk_alert.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job='disk-alert'` to the scraped data
```

## Step 5 — Restart Prometheus

```
systemctl restart prometheus
```



```
[root@ip-172-31-46-78 ~]# cd /opt/prometheus
[root@ip-172-31-46-78 prometheus]# vi disk_alert.yml
[root@ip-172-31-46-78 prometheus]# vi /opt/prometheus/prometheus.yml
[root@ip-172-31-46-78 prometheus]# systemctl restart prometheus
[root@ip-172-31-46-78 prometheus]#
```

## Step 6 — Verify Alert Rule

Open Prometheus UI:

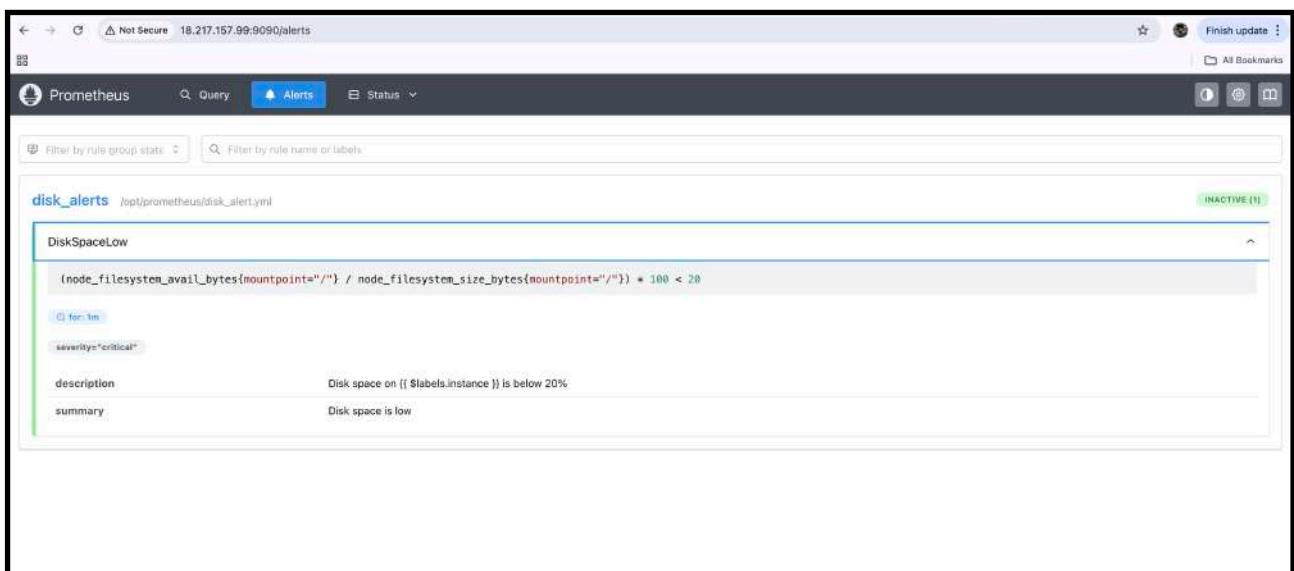
`http://PROMETHEUS_PUBLIC_IP:9090`

Go to:

Alerts

You should see:

DiskSpaceLow



Status will be:

Inactive

## Step 7 — Trigger the Alert (Target Node)

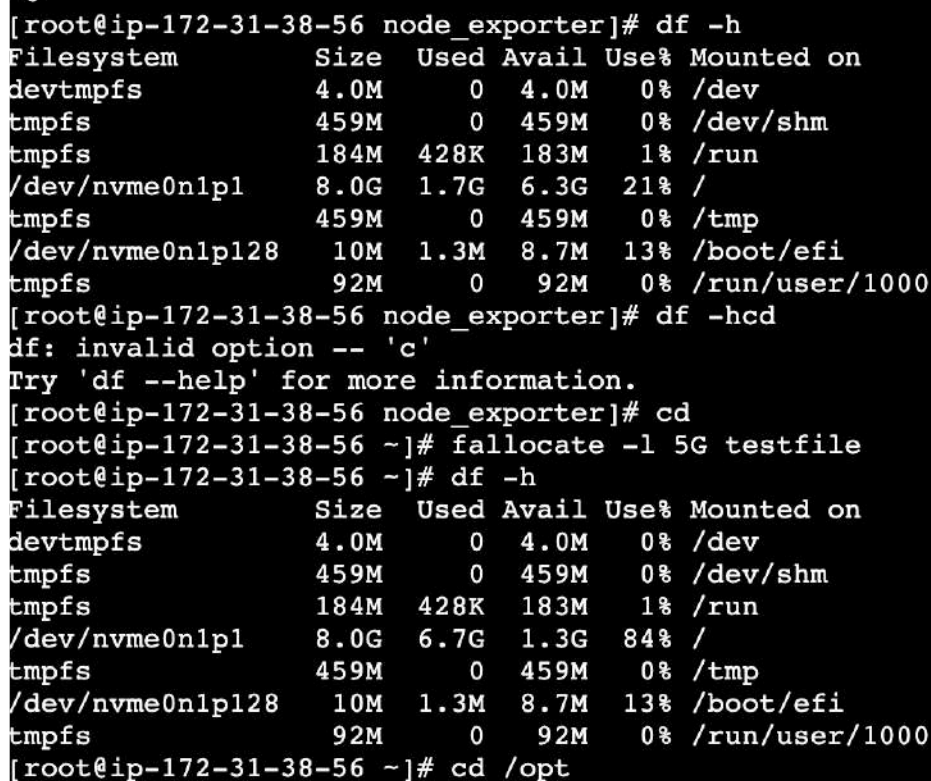
SSH into the **target node (promet-node-1)**.

Check disk usage:

```
df -h
```

Create a test file to increase disk usage:

```
fallocate -l 5G testfile
```



The screenshot shows a terminal session on a node named 'node\_exporter' with IP 172-31-38-56. It displays the output of the 'df -h' command twice. The first output shows the root filesystem at 21% usage. After running 'fallocate -l 5G testfile', the second 'df -h' output shows the root filesystem at 84% usage. The terminal also shows an attempt to use 'df -hcd' which fails with an invalid option error.

```
[root@ip-172-31-38-56 node_exporter]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0    4.0M  0% /dev
tmpfs           459M  0    459M  0% /dev/shm
tmpfs           184M  428K  183M  1% /run
/dev/nvme0n1p1  8.0G  1.7G  6.3G  21% /
tmpfs           459M  0    459M  0% /tmp
/dev/nvme0n1p128 10M   1.3M  8.7M  13% /boot/efi
tmpfs           92M   0    92M   0% /run/user/1000
[root@ip-172-31-38-56 node_exporter]# df -hcd
df: invalid option -- 'c'
Try 'df --help' for more information.
[root@ip-172-31-38-56 node_exporter]# cd
[root@ip-172-31-38-56 ~]# fallocate -l 5G testfile
[root@ip-172-31-38-56 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0    4.0M  0% /dev
tmpfs           459M  0    459M  0% /dev/shm
tmpfs           184M  428K  183M  1% /run
/dev/nvme0n1p1  8.0G  6.7G  1.3G  84% /
tmpfs           459M  0    459M  0% /tmp
/dev/nvme0n1p128 10M   1.3M  8.7M  13% /boot/efi
tmpfs           92M   0    92M   0% /run/user/1000
[root@ip-172-31-38-56 ~]# cd /opt
```

Example result:

Use% 84%

Now disk free space < 20%.

## Step 8 — Alert state change

Prometheus alert states:

Inactive → Pending → Firing

Because rule contains:

for: 1m

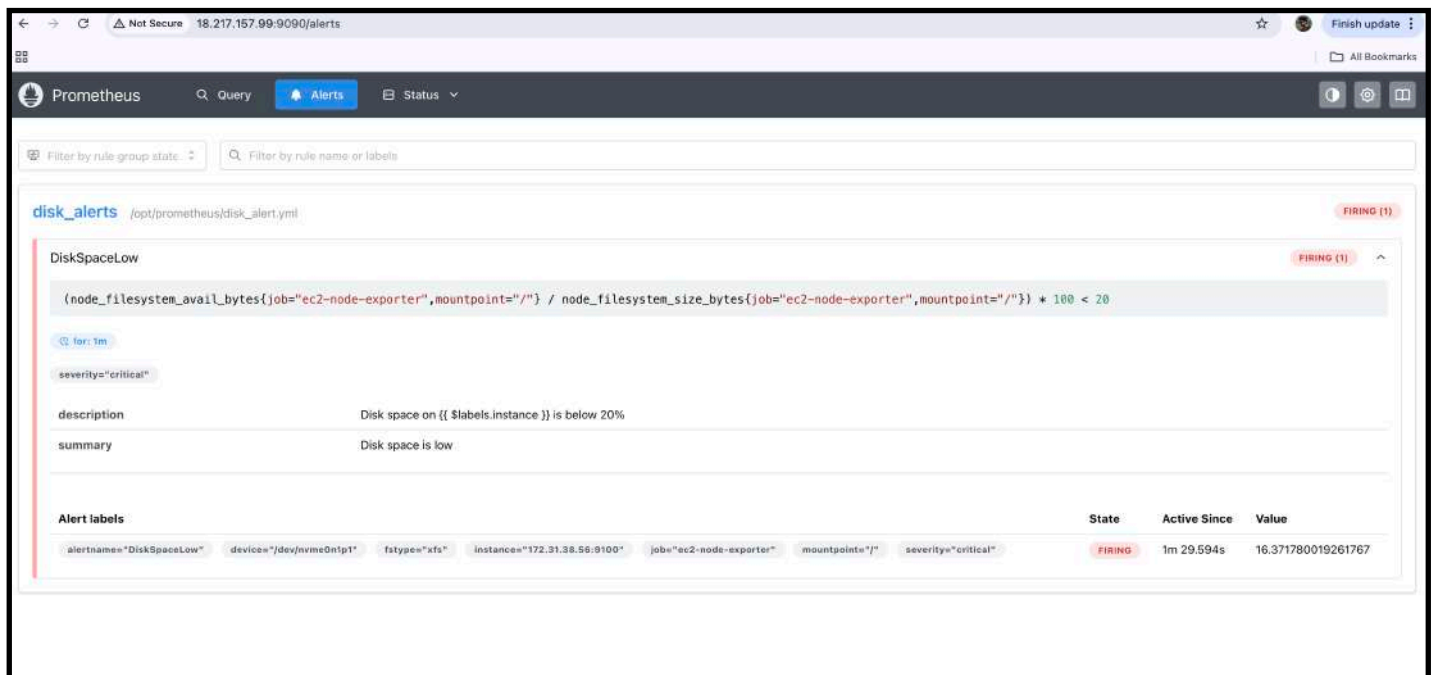
Prometheus waits **1 minute** before firing.

Check in UI:

[http://PROMETHEUS\\_SERVER\\_IP:9090/alerts](http://PROMETHEUS_SERVER_IP:9090/alerts)

You will see:

DiskSpaceLow → FIRING

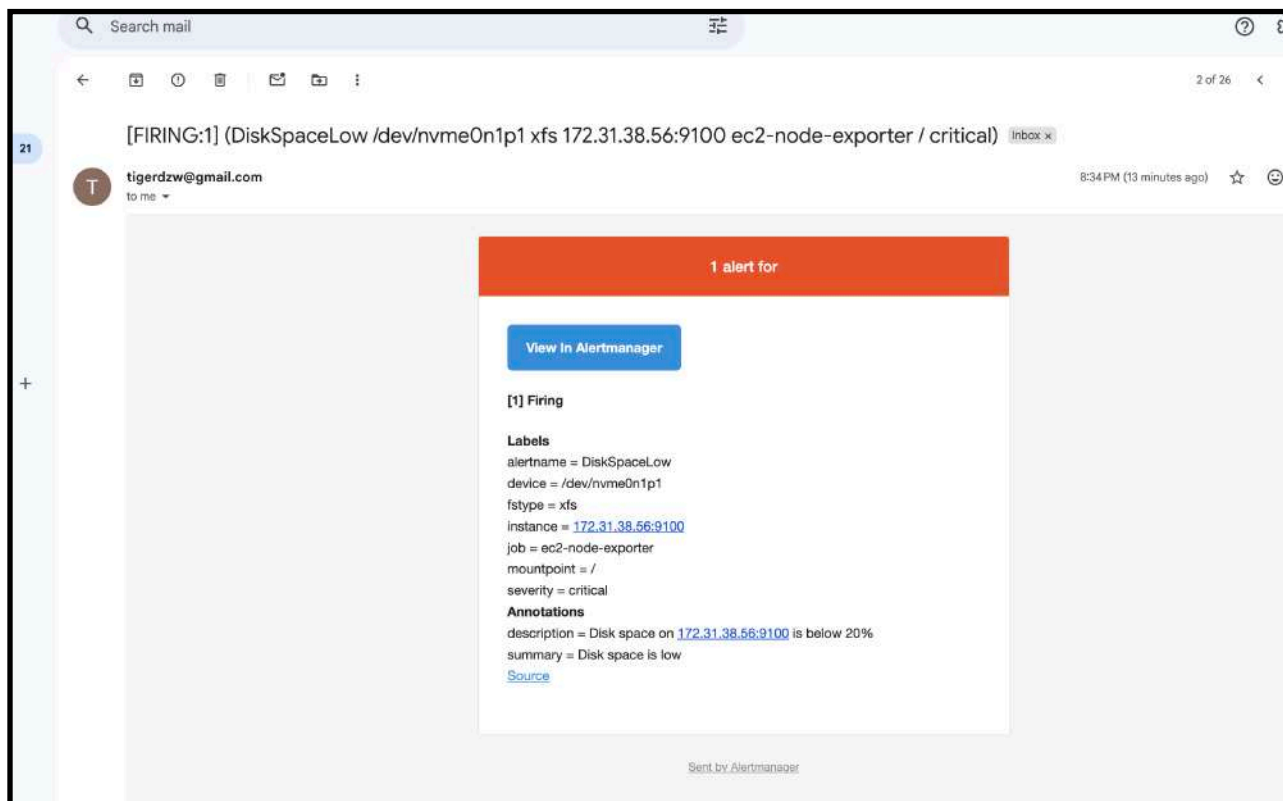


The screenshot shows the Prometheus Alerts UI in a web browser. The address bar displays `18.217.157.99:9090/alerts`. The Prometheus logo and navigation tabs (Query, Alerts, Status) are visible. The Alerts tab is active, showing a list of alerts. A filter bar at the top allows filtering by rule group state or name/labels. The alert list shows a single alert named `DiskSpaceLow` from the `/opt/prometheus/disk_alert.yml` file, which is in the `FIRING (1)` state. The alert details are expanded, showing the rule expression: `(node_filesystem_avail_bytes{job="ec2-node-exporter",mountpoint="/" } / node_filesystem_size_bytes{job="ec2-node-exporter",mountpoint="/"}) * 100 < 20`. The rule is configured with `for: 1m` and `severity: "critical"`. The description is "Disk space on {{ \$labels.instance }} is below 20%" and the summary is "Disk space is low". At the bottom, the alert labels are displayed as a horizontal list: `alertname="DiskSpaceLow"`, `device="/dev/nvme0n1p1"`, `fstype="xfs"`, `instance="172.31.38.56:9100"`, `job="ec2-node-exporter"`, and `mountpoint="/"`. A table below the labels provides further details:

| Alert labels                                                                                                                                                                                                                                     | State  | Active Since | Value              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|--------------------|
| <code>alertname="DiskSpaceLow"</code> <code>device="/dev/nvme0n1p1"</code> <code>fstype="xfs"</code> <code>instance="172.31.38.56:9100"</code> <code>job="ec2-node-exporter"</code> <code>mountpoint="/"</code> <code>severity="critical"</code> | FIRING | 1m 29.594s   | 16.371780019261767 |

# Step 8 — Alertmanager sends email

Alertmanager receives the alert and sends notification email.



Example receiver configuration:

```
receivers:  
- name: 'email-alert'  
  email_configs:  
  - to: 'tigerdzw@gmail.com'
```

You will receive email alert.

## Result

Prometheus successfully:

- ✓ Monitored disk usage
- ✓ Triggered alert when disk < 20%
- ✓ Sent email notification using Alertmanager

# Task 9 — Create Custom Grafana Dashboard (Prometheus)

## Step 1 — Open Grafana

Open Grafana in browser:

`http://GRAFANA_SERVER_IP:3000`

## Step 2 — Add Prometheus Data Source

In Grafana left menu:

`Connections → Data Sources`

Click:

`Add data source`

Choose:

`Prometheus`

Configure:

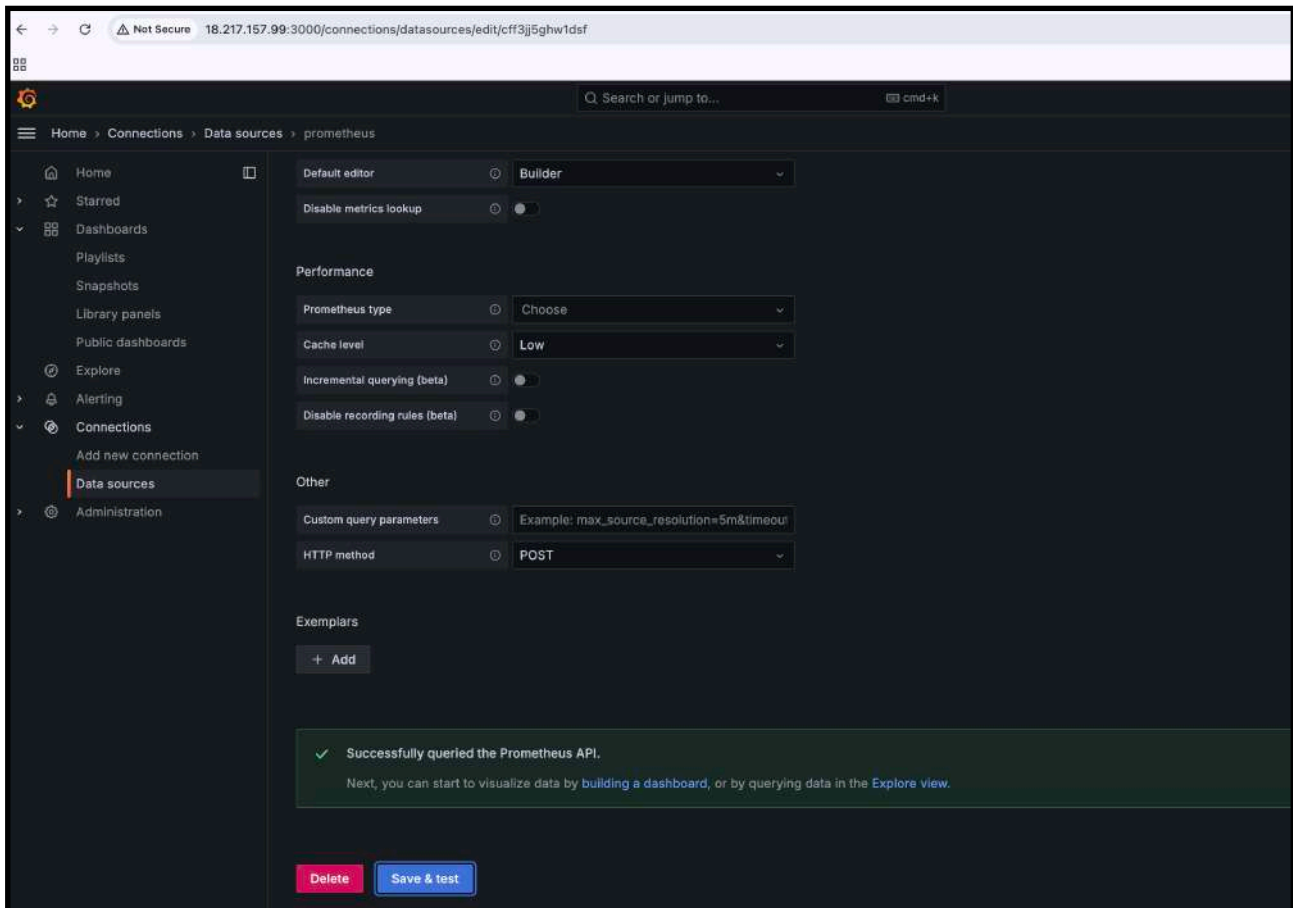
`URL = http://localhost:9090`

Click:

`Save & Test`

You should see:

`Data source is working`



## Step 3 — Create Dashboard

Left menu → click:

Dashboards

Then:

New → New Dashboard

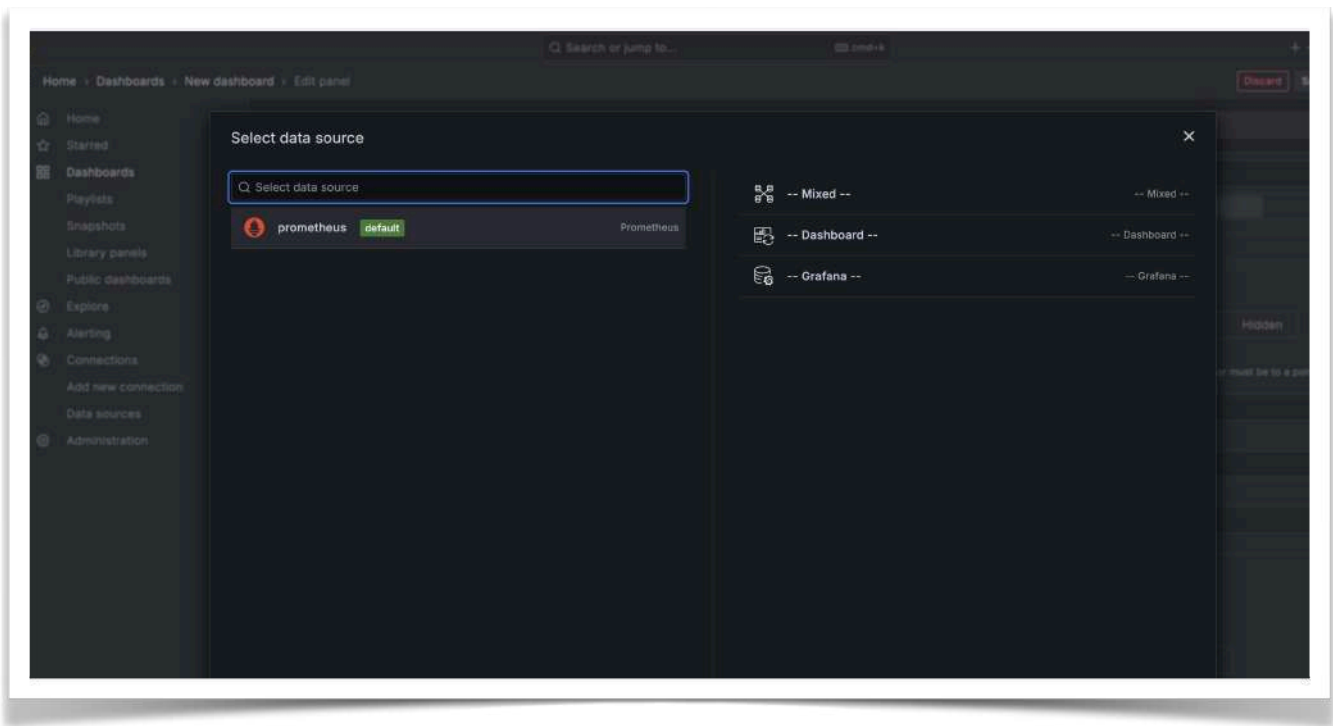
Click:

Add Visualization

Select:

Prometheus





## Step 4 — Add CPU Usage Panel

Query:

```
100 - (avg by(instance)
(irate(node_cpu_seconds_total{mode="idle"}[5m])) * 100)
```

Panel title:

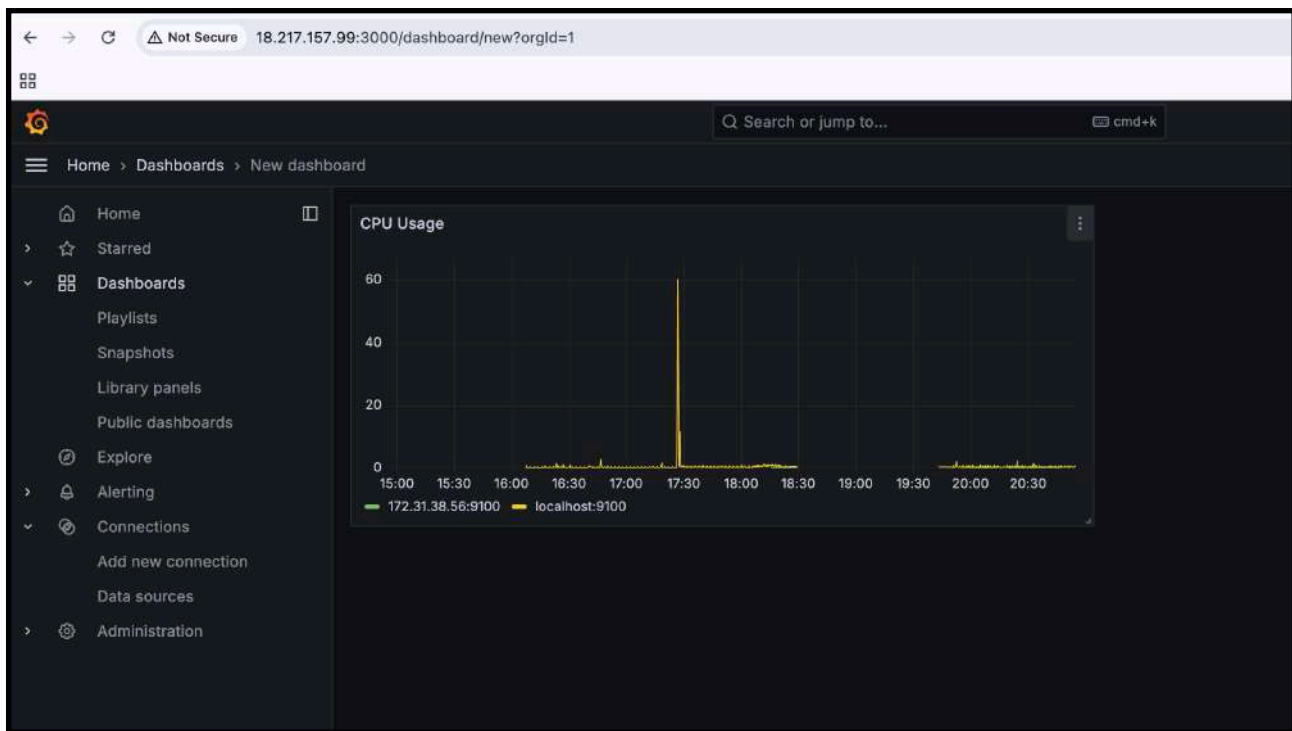
CPU Usage

Visualization type:

Time series

Click:

Apply



- `172.31.38.56:9100` → your **target EC2 node**
- `localhost:9100` → **Prometheus server node exporter**

So Grafana is correctly pulling metrics from **Prometheus**.

Now you just need to add **3 more panels** to complete the task.

## Panel 2 — Memory Usage

Click **Add visualization**.

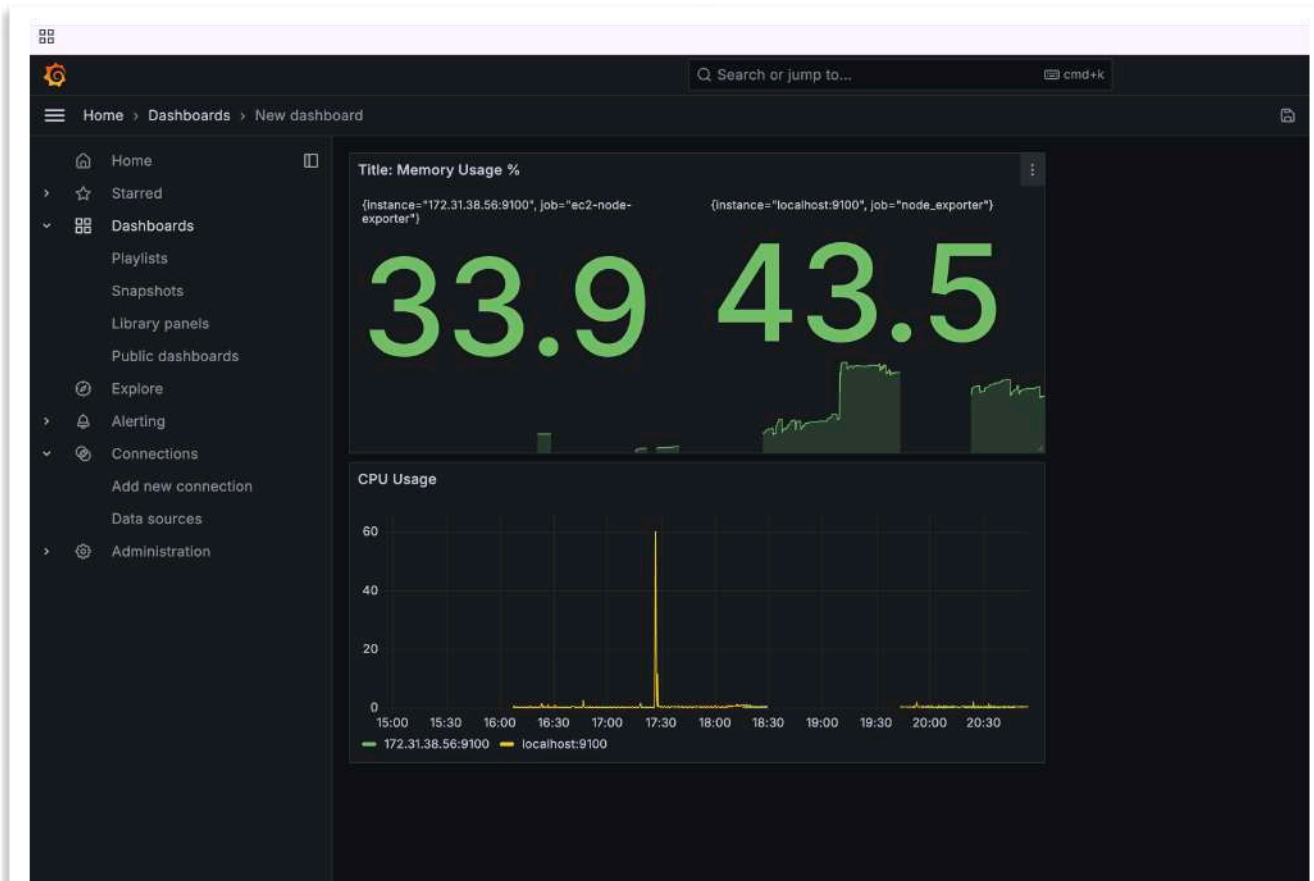
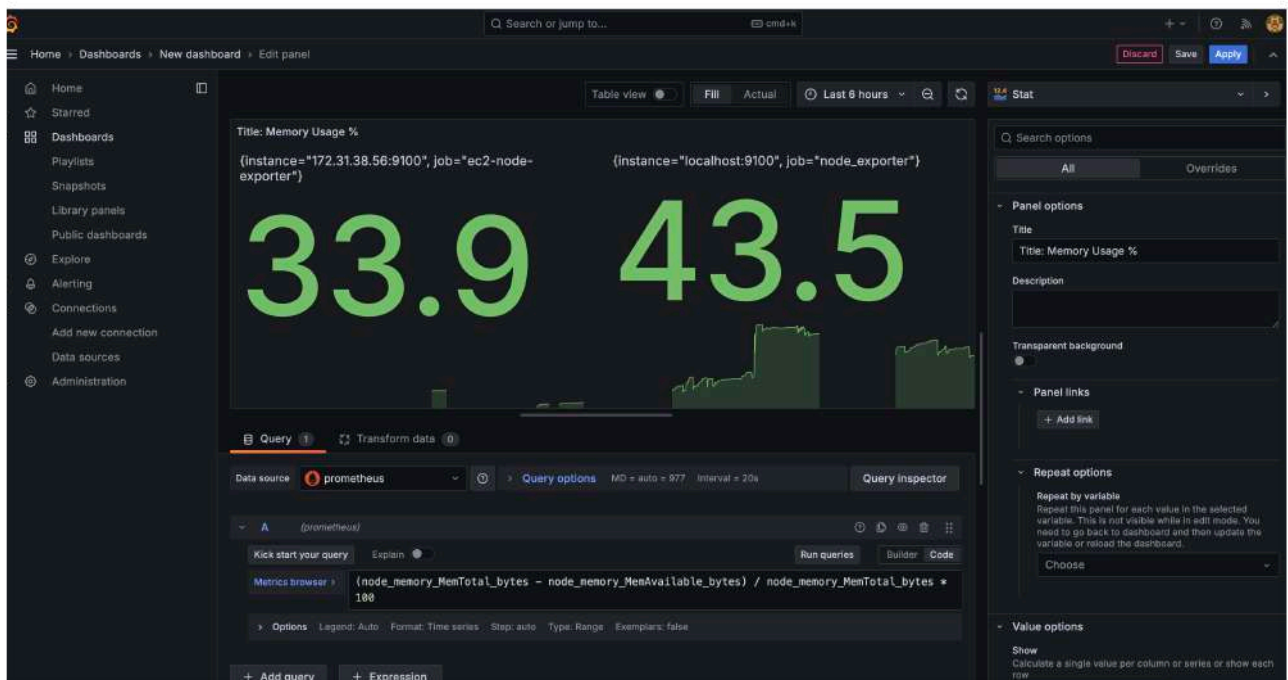
Query:

```
(node_memory_MemTotal_bytes - node_memory_MemAvailable_bytes) / node_memory_MemTotal_bytes * 100
```

Settings:

- Title: **Memory Usage %**
- Visualization: **stat**

Click **Apply**.



# Panel 3 — Disk Usage

Add another panel.

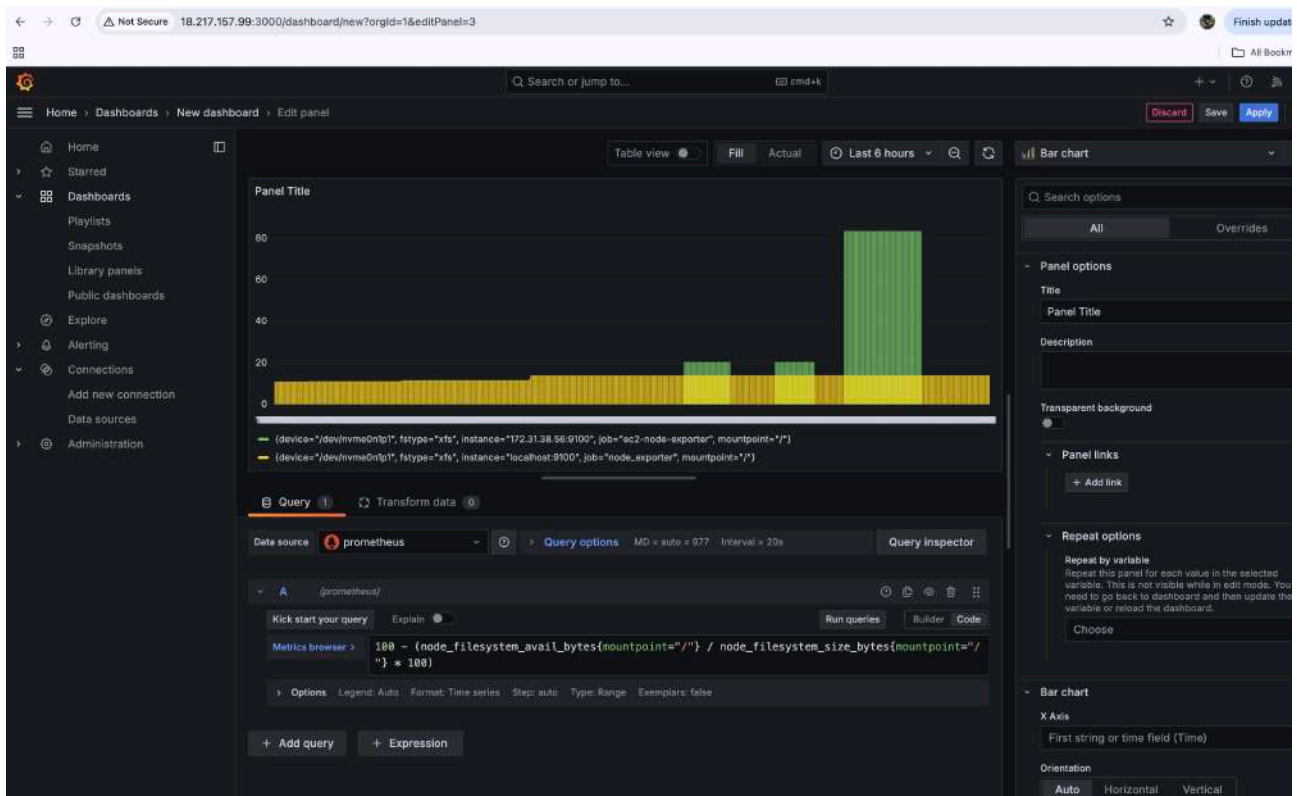
Query:

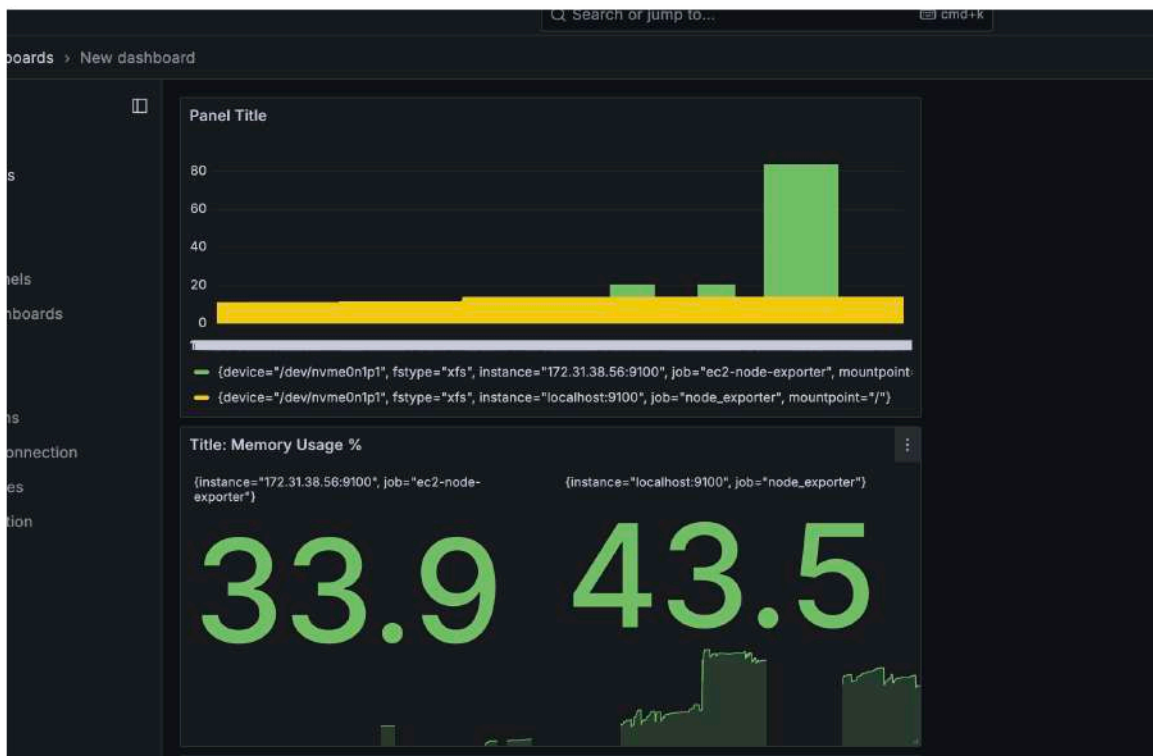
```
100 - (node_filesystem_avail_bytes{mountpoint="/" } /  
node_filesystem_size_bytes{mountpoint="/" } * 100)
```

Settings:

- Title: **Disk Usage %**
- Visualization: **bar chart**

Click **Apply**.





## Panel 4 — Server Uptime

Add panel.

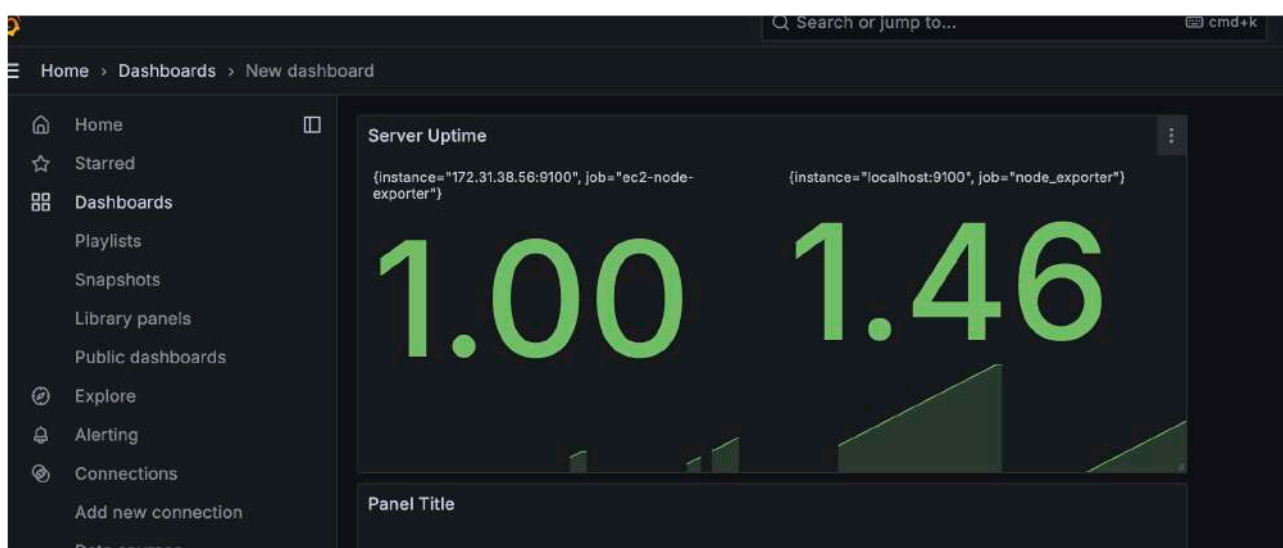
Query:

```
node_time_seconds - node_boot_time_seconds
```

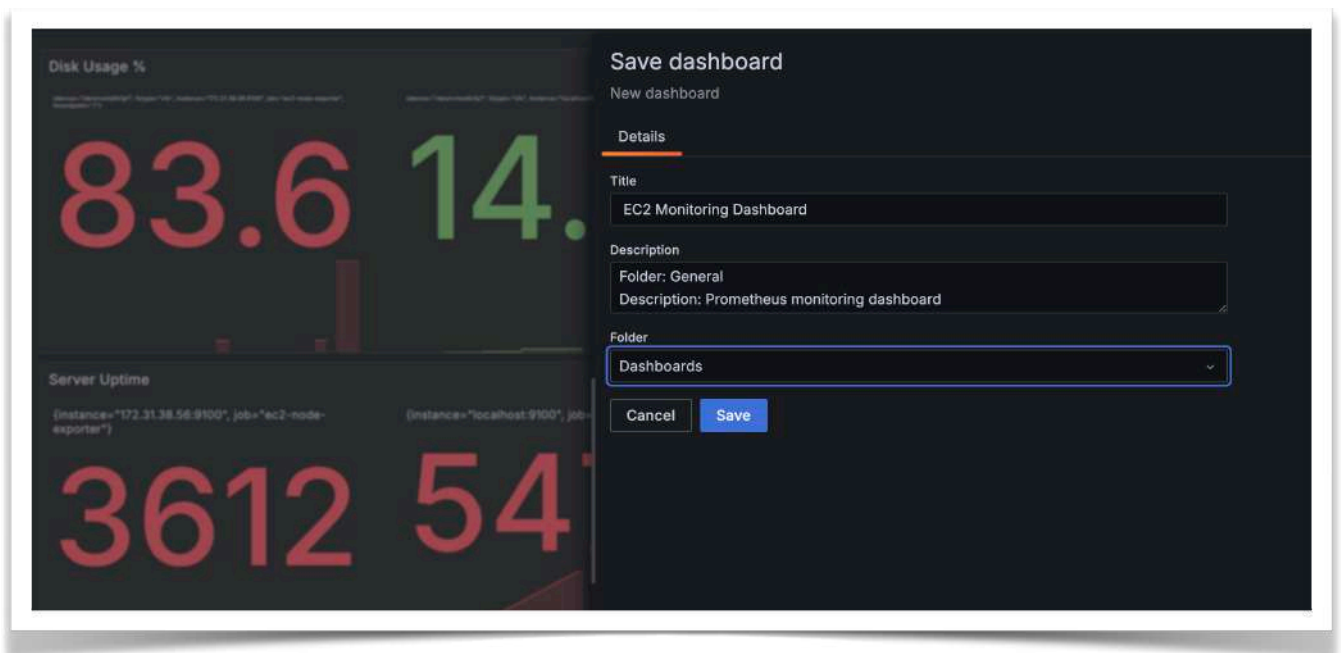
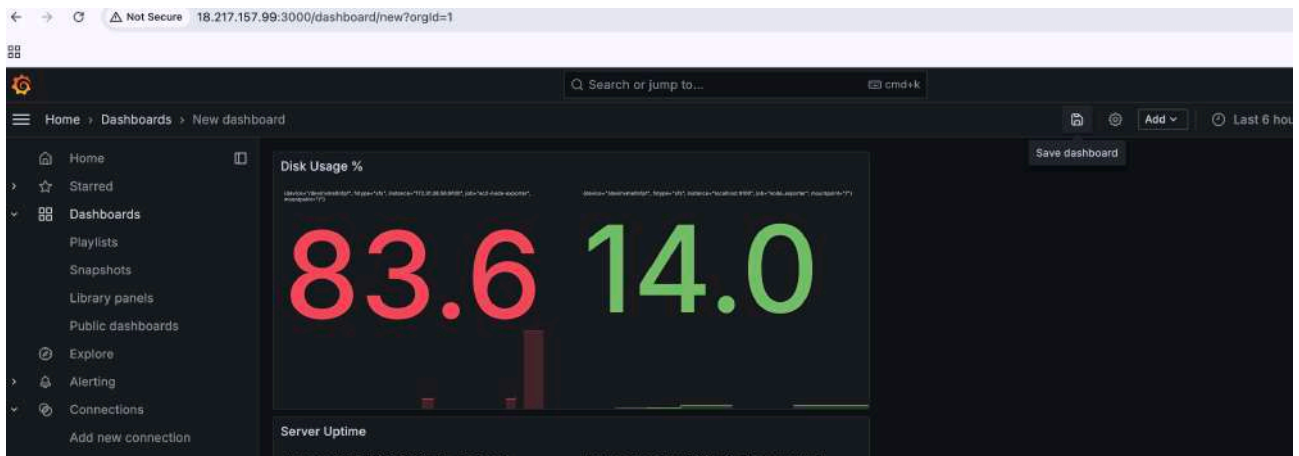
Settings:

- Title: **Server Uptime**
- Visualization: **Stat**

Click **Apply**.



—>now click on save dashboard in right side up



**Click Save**

Click the **Save** button.

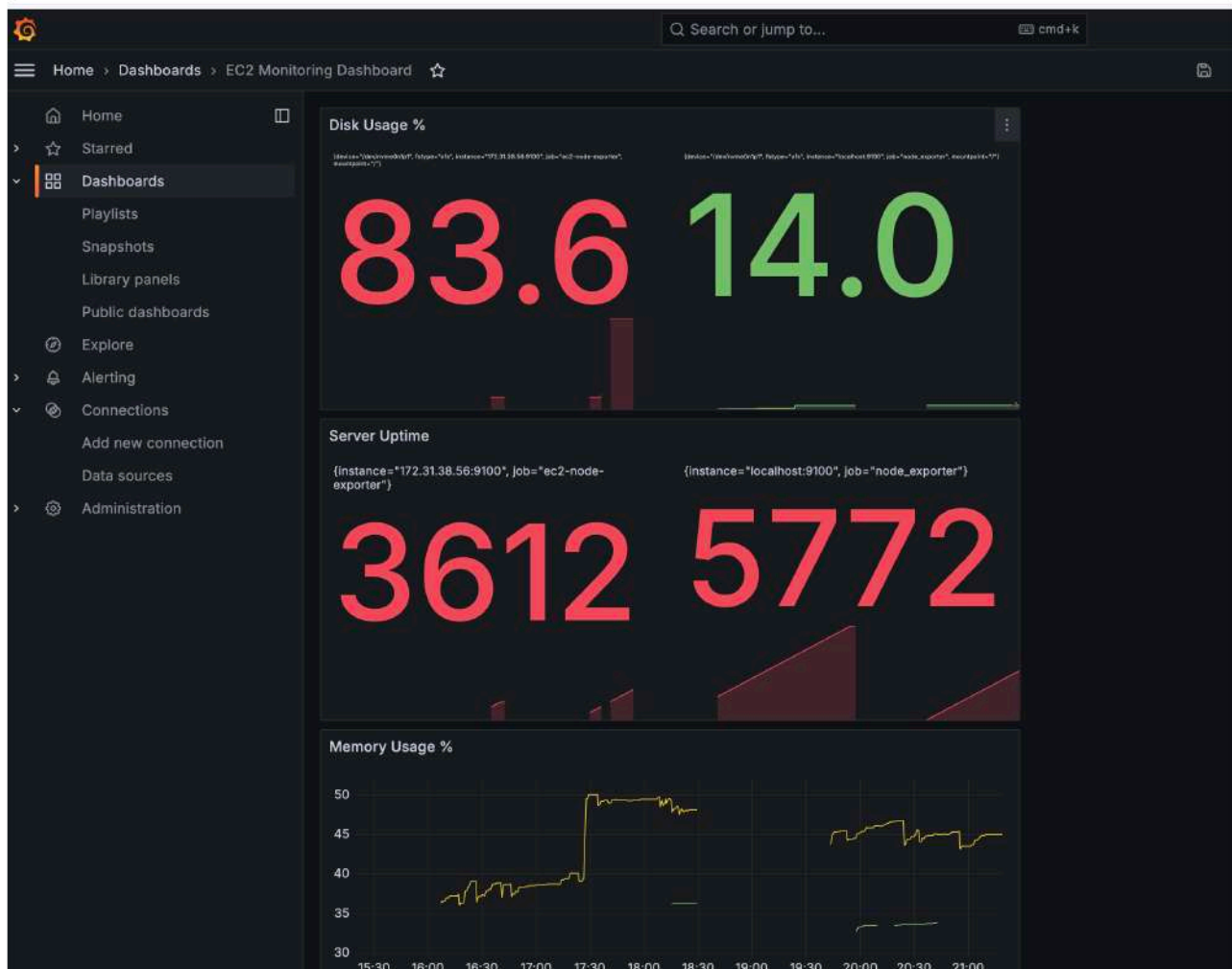
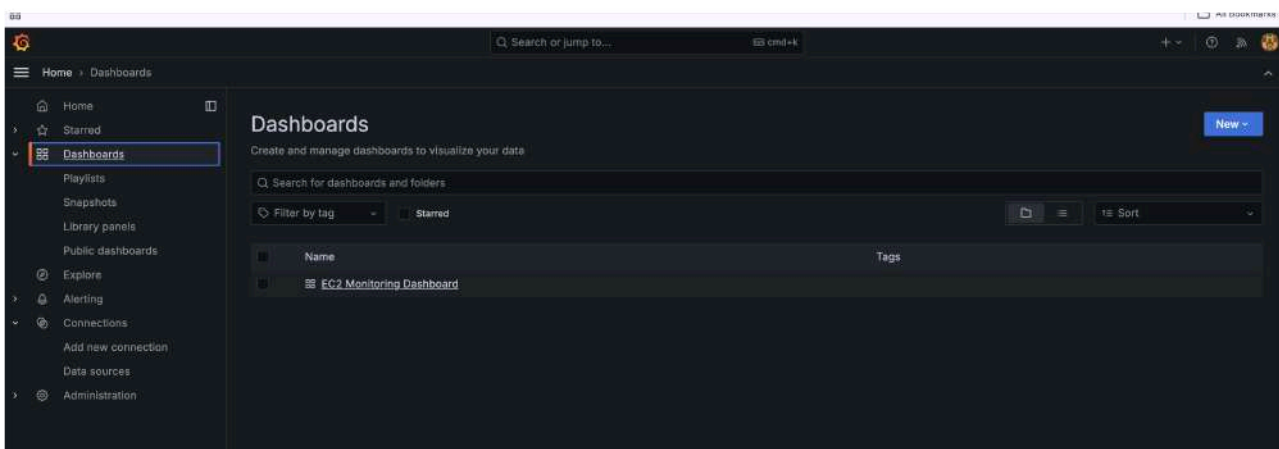
## Verify

Now go to the left menu:

Dashboards → Browse

You will see:

EC2 Monitoring Dashboard





Your **custom Grafana dashboard** is now saved.

I implemented a monitoring system using Prometheus and Grafana. Node Exporter collects system metrics from EC2 instances. Prometheus scrapes the metrics and Alertmanager sends email alerts when disk space falls below 20%. Grafana visualizes CPU, memory, disk usage, and server uptime through a custom dashboard.

## Your Completed Stack

Node Exporter



Prometheus



Alertmanager → Email Alerts



Grafana Dashboard

